

Sources of resistance to viruses of pepper (*CapsiCule* spp.): a catalog

S., K., Green and I.S. Kim

*Asian Vegetable Research and Development Center
Taiwan, ROC*

*Agricultural Sciences Institute, Rural Development Administration
Republic of Korea*

1 /4702

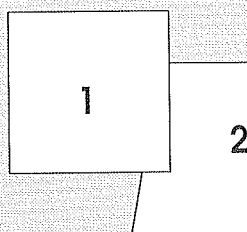


Asian Vegetable Research and Development Center

© 1994. Asian. Vegetable Research and Development Center
P.O. Box 205, Taipei 110099

Green, S.K. and Kim, J.S. 1994. Sources of resistance to viruses of pepper (*Capsicum* spp.): a catalog. Asian Vegetable Research and Development Center. Technical Bulletin No. 20, 72 p.

The preparation of this publication was partly funded by the Ministry of Economic Cooperation/German Agency for Technical Cooperation (BMZ/GTZ).



About the icons: 1 — virus-infected pepper plant
2 — healthy pepper plant
(Photos by Chen Ming-che, AVRDC)

AVRDC Publication 94-427
ISBN: 92-9058-087-9

technical editing: Katherine Lopez

Contents

Acknowledgments	4
Introduction	5
Chilli veinal mottle virus	7
Cucumber mosaic virus	10
Curly top virus	20
Leaf curl virus	21
Pepper mottle virus	25
Pepper veinal mottle virus	27
Pepper vein banding virus	28
Potato virus X	29
Potato virus Y	31
Tobacco etch virus	41
Tobacco ring spot virus	45
Tobamoviruses	46
Tomato spotted wilt virus	57
Unspecified viruses	58
Appendix 1: Viruses reported to infect pepper	59
References	63

Acknowledgments

The authors would like to thank the following researchers for providing information for this bulletin:

L.L. Black, Louisiana State University, USA; P. Bosland, New Mexico State University; B.H. Chew, Malaysian Agricultural Research and Development Institute (MARDI), Malaysia; J. Horvath, University of Agricultural Sciences, Keszethely, Hungary; A. Moor, Vegetable Crops Research Institute, Budapest, Hungary; H. Nagai, Instituto Agronomico de Campinas, Campinas, Brazil; A. Palloix and E. Pochard, Institut National de la Recherche Agronomique (INRA), Montfavet, France; J. Singh, Punjab Agricultural University, Ludhiana, India; and Y.J. Yoon, Horticultural Experiment Station, Suwon, Korea.

The authors also thank B.H. Chew, J.M. Poulos (AVRDC), and J. Singh for reviewing the manuscript and for making valuable suggestions to improve it.

This publication was partly funded by the Ministry of Economic Cooperation/German Agency for Technical Cooperation (BMZ/GTZ).

Introduction

Some 35 viruses have been reported to infect peppers (*Capsicum* spp.) (Appendix 1). Many of these occur worldwide and cause considerable economic losses in terms of reduced quality and marketable yield. For detailed information concerning viruses of peppers the reader is referred to the AVRDC Technical Bulletin No. 18, Characteristics and control of viruses infecting peppers: a literature review. Control of these viruses by chemicals or cultural management practices has not been very effective. The planting of resistant cultivars offers the best way to control these viruses. A number of sources of virus resistance have been reported in wild and cultivated lines and several virus-resistant lines have already been released. Among these are lines with resistance to potato virus Y (PVY), tobacco etch virus (TEV), pepper mottle virus (PeMV), and certain tobamoviruses. However, other viruses, particularly cucumber mosaic virus (CMV) and the leaf curl virus complex, continue to pose a challenge to breeding programs.

Information on sources of resistance to many viruses of peppers is scattered and is often contained in progress reports or similar publications with a limited circulation. Furthermore, some reports are written in languages not widely understood.

The primary purpose of this publication is to provide a practical reference for breeders, pathologists, and agricultural specialists working on virus diseases of peppers and to encourage the evaluation and use of germplasm reported to carry virus resistance or tolerance. Sufficient progress has been attained in the development of germplasm resistant to viruses to make this tactic a very promising approach to control. Yet there is evidence that breeders in developing countries are not using the available resistant cultivars and that many are not sufficiently aware that these lines exist.

Information presented here may not be complete. The literature search tried to cover as much of the available published information as possible. We also contacted researchers of national programs with active pepper breeding programs to obtain additional information on sources of resistance. We hope that important omissions as well as results of future research will be brought to our attention.

Commercially available sources of resistance are generally not included as they should be easily accessible from the various seed catalogs. The information presented here is grouped alphabetically by individual viruses. Within each virus section the references are listed in chronological order. We have attempted to include additional information on the type of resistance wherever it was available since some of the resistant lines have been identified and studied by various researchers under different conditions. We avoided grouping these together but, instead, listed these separately to provide as much information as possible. The terminology for resistance, immunity, and tolerance is indicated verbatim from the cited reference. Similarly, the nomenclature for the various *Capsicum* species is mostly that used by the original authors.

Some of the virus-resistant germplasm have been assembled by AVRDC and can be freely obtained from the center. Virus-resistant germplasm can also be obtained from the International Plant Genetic Resources Institute (IPGRI)-designated global pepper germplasm collection at the Centro Agronomico Tropical de Investigacion y Pnsenanza (CATIE), Costa Rica. For each resistance source the original authors are listed and, thus, they may also be contacted,

We thank the many people who furnished information for inclusion in this list, and we hope to obtain feedback from the readers on resistance sources we may have omitted as well as new sources of resistance to update this catalog in the future.

S.K. Green and J.S. Kim

Chilli Veinal Mottle Virus (CVMV)

References

1. *Capsicum annuum* HDA 832, HDA 248 AVRDC 1990a
2. *C. annuum*. HDA 832, Serrano Huasteco AVRDC 1990b
3. *C. annuum* VC 33a, VC 37a, VC 58, BG1, Chili Piquin, Erectar, Serrano San Luis, Taiwan 1, Tiwari, Thailand
Varieties FCM 2, FCM 3, FCB 1, FCB 3, DCM 4, DC, DCB 1

C. annuum. VC 17a, VC 40a, VC 41a, Lorai, Mexico, Perennial, Punjab Lal, Serrano Tampiqueno, Szechuan
— also resistant to CMV
4. *C. annuum* HDA 210bis, HDA 230, HDA 252, HDA 295 AVRDC 1991

C. annuum HDA 248, HDA 249
C. frutescens PSP-11
— also resistant to PVMV

C. annuum. HDA 273
— also resistant to PVY

C. minimum Perennial HDV, HDA 832
— also resistant to PVMV and PVY
5. *C. annuum* Lorai Singh 1992
- also resistant to TMV, CMV, PVY, and leaf curl virus

C. annuum Punjab Lal
— also resistant to TMV, CMV, and leaf curl virus
6. *C. annuum* Cili Padi No. 6 Chew 1993
resistance to CVMV is controlled by a pair of recessive genes (possibly a number of independent resistance genes are involved)

C. annuum Serrano Huasteco
resistance to CVMV is controlled by a pair of recessive genes (possibly a number of independent resistance genes are involved)
also resistant to CMV

Chilli Veinal Mottle Virus

References

- C. annuwn* Avelar, AVRDC Acc. C 011751-1 (Criollo de Morelos 334), C 01664, PBC 37, BG-1, Lorai, Osir (F₁), Pant C-1, Serrano Huasteco, Serrano Tampiqueno, Tiwari II AVRDC 1993
- C. annuunt*. Perennial
resistant to CVMV isolates from Taiwan, Malaysia, and Thailand
also resistant to a Malaysian CMV isolate
- C. a;iuu nt* Perennial HDV
resistant to CVMV isolates from Malaysia, Taiwan, and Thailand
also resistant to CMV isolates from Malaysia and Thailand
- C. annuunt* HDA 230, HDA 248, HDA 249, HDA 252, Seagochu
— resistant to CVMV isolates from Thailand and Malaysia
- C. annuunt* HDA 295
— resistant to CVMV isolates from Thailand, Malaysia, and Taiwan
— also resistant to an isolate of CMV in Malaysia
- C. aim* = HDA 210 bis
— resistant to CVMV isolates from Malaysia and Taiwan
— also resistant to a Thailand isolate of CMV
- C. annuun* HDA 832, Serrano Huasteco
— resistant to CVMV isolates from Taiwan and Malaysia
— also resistant to PVMV
- C. frutescens* PSP-11
— resistant to CVMV isolates from Taiwan and Malaysia
- C. annuunt* HDA 273
— resistant to CVMV isolates from Malaysia and Taiwan
— also resistant to CMV isolates from Malaysia, Taiwan, and Thailand

Chilli Veinal Mottle Virus	References
8. <i>C. ain uunr</i> PBC 092, PBC 180, PBC 349, PBC 370, PBC 495, PBC 522, PBC 523, PBC 545, P13C 549, PBC 569, PBC 691, PBC 787, AVRDC Acc. C 00563, C 01315	AVRDC 1994
<i>C. anuum</i> AVRDC Acc. C 01664 — also resistant to PVY	

Cucumber Mosaic Virus (CMV)

References

- | | | |
|----|---|---|
| 1. | <i>C. frutescens</i> LP-1
— tendency to escape infection;
less concentration of CMV within plant
CMV resistance controlled by a single
recessive gene
also resistant to TEV
(resistance to TEV is dominant) | Barrios et al. 1971 |
| 2. | <i>C. annuum</i> Antibois, Piment Sucette
CMV-tolerant; tolerance associated with
small fruit size
white fruit-type peppers have some
tolerance to CMV | Pochard and Chambonnet 1972
Lovisolo and Conti 1976
Zatyko 1978 |
| 3. | <i>C. baccatum</i> var. <i>pendulum</i> 3-4
resistant to virus migration
the virus is localized in lateral shoots, the
invaded tissue becoming necrotic (a form of
hypersensitivity); inhibitors may also be present
also resistant to PVY | Pochard and Chambonnet 1972
Pochard 1977
Palloix et al. 1989 |
| | <i>C. annuum</i> A6 (derived from Val x Babas),
Babas (derived from <i>C. annuum</i> Bastidon x
<i>C. baccatum</i> x Val), Val | |
| | <i>C. frutescens</i> LP 1
— has tendency to escape infection | |
| | <i>C. annuum</i> Avelar, Ikeda, Moura
— low level of infection in the field | |
| 4. | <i>C. annuum</i> (8 lines)
- resistant to CMV | Saccardo 1973 |
| | <i>C. annuum</i> (13 lines)
— tolerant to CMV | |
| | <i>C. baccatum</i> var. <i>microcarpum</i> (1 line)
— resistant to CMV | |
| | <i>C. baccatum</i> var. <i>pendulum</i> (1 line)
— resistant to CMV | |

Cucufnber Mosaic Virus (CMV)

References

- C. chinense* (1 line)
— resistant to CMV
- C. frutescens* (10lines)
— resistant to CMV
5. *C. annuum*. (8 wild lines) Saccardo 1974
C. baccatuni var. *pendulnm*. ("1 line)
C. baccatuni var. *microcarpuut*. (1 line)
C. chinense (1 line)
C. frutescens (10 lines)
6. *C. annuum* Perennial, S 5-4, S 20-1, S 41-1, S 118-2 Singh and Thakur 1977
 - single recessive gene (*cm*) for resistance to CMV
 also possess single recessive gene (*lc*) for TLCV
 also possess single recessive gene(*l*) for TMV (strain 1);
 (According to Pochard et al. 1983, there is no resistance to TMV pathotype 0, 1-2, and 1-2-3 in Perennial)
 the 3 separate genes are linked
 (crossover value 38-40%)
7. *C. annuum* Kalocsai VI Szirmai 1978
 (derived from a cross between a half wild hypersensitive Chinese variety and the susceptible old variety Szegedi 91-95)
 — CMV-tolerant
 - also TMV-resistant
8. *C. annuum* EC 409511, S 4-5, S 5-3, S 5-4, S 6-3, S 6-6, S 6-8, S 11-1, S 11-4, S 20-1, S 34-7, S 41-1, S 41-2, S 72-4, S 118-2, S 118-4, Perennial Singh and Thakur 1979
- C. annuum* Bangla Green, HS-1, Jamni, Laichi, Longi, Lorai, Pant C-1, Perennial, Tiwari, S 20-1, S 118-2
 — also resistant to TMV
 - also field resistant to leaf curl virus
9. *C. annuum*. Goliat (Magician) Zatyko et al. 1979
 - CMV-tolerant
- C. annuum* Taltos
 - CMV-tolerant
 - CMV tolerance derived from *C. annuum* Antibois

Cucumber Mosaic Virus (CMV)

References

10. Unidentified local Indian variety
 - immune to CMV and PVX
 - tolerant to TMV and leaf curl virus

C. annuum Pant C-1, Pant C-2

 - tolerant to CMV, PVX, TMV, and leaf curl virus

C. annuum. NP 46-0, X-196

C. frutescens EC 31352

 - tolerant to CMV and PVX

C. common var. *angulosum*

 - tolerant to CMV
 - also tolerant to PVX
11. *C. autumcm.* Ikeda, Moura
 - resistant to aphid transmission
12. Some species (among 48 plants tested)
13. *C. annuum* var. *minimuum*. Rama
 - dominant gene (*Riv*) for reduced accessibility of the leaves to CMV by mechanical inoculation

C. annuum Perennial

 - recessive genes for reduced accessibility of the leaves to CMV by mechanical inoculation

C. annuum PM 642, PM 660, PM 662

 - partly resistant to CMV
14. *C. annuum* S 38-2-1, S 94-4-9-3, S 101-2-33
 - tolerant to CMV
 - also tolerant to PVX, TMV, PVY, and leaf curl virus
15. *C. annuum* X-196, X-197
 - tolerant to PVX
16. *C. annuum* Delhi. Local
 - immune to CMV and PVX
 - tolerant to TMV and leaf curl virus

Konai and Nariani 1980

Cohen 1982

Pasco et al. 1982

Pochard 1982

Tewari and Viswanath 1982

Konai and Nariani 1983

Konai and Nariani 1983
 Tewari and Viswanath 1986
 Tewari 1986

C. ammnn. Pant C-1, Pant C-2
 — tolerant to CMV, PVX, leaf curl virus, and TMV

Cucumber Mosaic Virus (CMV)

References

- C. annuum* X-196, X-197
— tolerant to CMV
17. *C. annuum* Philomele "1.
(derived from *C. baccatum* var. *pendulum* 3-4)
— field resistant to CMV
18. *C. annuum* HDA 249, HDA 260, HDA 268, HDA 293
(androgenetic doubled haploids from the cross
Perennial x Yolo Wonder)
— highly tolerant to CMV under field conditions
and by artificial inoculation
- C. annum.* HDA 206, HDA 211, HDA 227, HDA 230,
HDA 232, HDA 234, HDA 242, HDA 248, HDA 252,
HDA 255, HDA 258, HDA 265, HDA 267, HDA 283,
HDA 294, HDA 295
— CMV-tolerant
19. *C. annuum.* Lorai, Longi, Pant C-1, Perermial, S 118-2
resistant/tolerant to CMV and TMV under
field conditions, showing very mild to
moderate symptoms
also resistant/tolerant to leaf curl virus
20. *C. annuum* PI 286419
virus localized in inoculated leaves
resistant only to Massachusetts CMV isolates
not to California isolates
21. *C. pubescens*
— also resistant to PVY and ToMV
22. *C. baccatum* var. *microcarpmr*
C. chacoense
C. frutescens var. *cerasiforme*
C. frutescens Tabasco
— also resistant to TMV
- C. baccatum* var. *pendulum*
— also resistant to TMV and ToMV
23. *C. baccatum* var. *pendulum* 3-4
— partial resistance to CMV
- Marchoux et al. 1983
- Pochard et al. 1983
- Sharma et al. 1983
Sharma and Singh 1985
Singh 1992
- Cuevas and Nicklow 1985
- Horvath 1986a
- Horvath 1986b
- Miladinovic et al. 1986

Cucumber Mosaic Virus (CMV)

References

- Lines 12/84, 14/84
 reaction to CMV similar to
C. baccatum var. *pendulum* 3-4
24. *C. annuum* Perennial
 resistant to virus multiplication
 (polygenic recessive inheritance) Pochard et al. 1986
 Pochard and Daubeze 1989
25. *C. annuum* var. *minimum* Rama Pochard et al. 1986
 (wild type from Nicaragua)
 — resistant to infection
 — 1 single dominant gene
- C. annuum*. PM 815 (Er-Fu-Tou),
 PM 819 (Chao-Tian-Jiao) (both of Chinese origin)
 resistant to virus multiplication
26. *C. annuum* Jangsu (F₁ hybrid) RDA 1986
 (CMV resistance derived from a Thailand local
 variety Taigukgochu)
 — also resistant to TMV
- C. annuum* Wongkyo 306
 (CMV resistance derived from Taigukgochu)
 — also resistant to TMV
27. *C. annuum* Bangla Green, HS-1, Januli, Laichi,
 Longi, Lorai, Pant C-1, Perennial, S 20-1, S 118-2
 (released as Punjab Lal in 1985), Tiwari Singh and Kaur 1986
 Singh 1992
 — also field resistant to leaf curl virus
 — resistant to CMV and TMV
28. *C. annuum* S 38-2-1, S 96-4-9-3, S 101-2-33 Tewari and Viswanath 1986
 — tolerant to CMV
 — also tolerant to TMV, PVX, and leaf curl virus
- C. annuum* S 38-3-19, S 42-2-4
 — tolerant to CMV
 — also tolerant to PVX and leaf curl virus
- C. annuum* S 52-1-6, S 96-4-8
 — tolerant to CMV
 — also tolerant to leaf curl virus

Cucumber Mosaic Virus (CMV)

References

29. *C. annuunt* Sel 52-18
(derived from a cross of Pusa Jwala and Delhi Local)
— tolerant to CMV
— also tolerant to leaf curl virus and potato virus X
Tewari 1986
30. *C. pubescens* LS 1659
— plants of this line show only very mild symptoms and are not damaged
Narikawa et al. 1987
31. *C. annuun* MRCH
— tolerant to CMV
resistance is polygenic and linked to small fruit size
Shifriss and Cohen 1987
32. *C. annuum* Bhivapur 1, Pantnagar-1,
Pantnagar-2, PKV-2, Sabouragar, Surjamani
Peshney and Moghe 1988
- C. annuum* Perennial
— immune to CMV
- also immune to ToMV
- C. annuum* G5, KI, CA 1068, PKV 1
moderately resistant to CMV
also moderately resistant to TMV
- C. frutescens* Lawangi
— moderately resistant to CMV
— also resistant to TMV
33. *C. annutnn* Perennial
partially resistant to CMV
(reduced virus multiplication)
highly tolerant to CMV under field conditions
delayed symptom appearance, mild symptoms
also partly resistant to PVY (pathotypes 0, 1, 1.2)
Pochard 1988
(personal communication)
Pochard and Daubeze 1989
- C. annuunt* HDA 201, HDA 210 bis, HDA 230,
HDA 248, HDA 249, HDA 252, HDA 260, HDA 268,
HDA 273, HDA 295
— field tolerant to CMV
- also resistant to PVY (0, 1, 1.2)
- tolerant to TYLCV under field conditions

Cucumber Mosaic Virus (CMV)

References

-
- | | |
|---|---|
| <p>34. <i>C. baccatum</i>, PEN 3-4
 partially resistant (resistant to migration)
 only necrotic reaction of the invaded tissues;
 virus does not move to axillary shoots in 5-leaf
 stage plants inoculated on the 3rd leaf and then
 decapitated above the 3rd leaf; however, when
 inoculated at the cotyledon stage, some plants
 become invaded by CMV</p> <p>35. <i>C. baccatum</i> var. <i>pendulum</i> LS 390
 — tolerant to CMV in Japan</p> <p><i>C. annuum</i> 505
 — limited virus migration within the plant</p> <p>36. <i>C. annuum</i> Milord
 — partially resistant (reduced virus migration;
 CMV is localized in tissues on the side of the
 inoculated leaf)
 resistance is suppressed at low temperatures
 (12–25°C)</p> <p><i>C. annuum</i> Ivan
 — resistant to CMV
 — virus remains localized in the inoculated leaf</p> <p><i>C. annuum</i> PM 815 (Er-Fu-Tou)
 — tendency to recover from viral infection</p> <p>37. <i>C. frutescens</i>
 (several wild accessions from Latin America)</p> <p>38. <i>C. annuum</i> Punjab Lal
 also resistant to PVY, TMV, and leaf curl virus</p> <p>39. <i>C. annuum</i> Bangla-Green 1 (BG-1), Punjab Lal (PL)
 reduced CMV multiplication
 resistance accompanied by increase in total
 and orthodihydroxy phenols and high levels
 of quinones after virus infection</p> <p>40. <i>C. baccatum</i>
 <i>C. pubescens</i>
 — more resistant than commercial cultivars</p> | <p>Dufour et al. 1989
 Palloix et al. 1989
 Chaîne et al. 1992</p> <p>Cited by Kostova et al. 1989</p> <p>Pochard and Daubeze 1989
 Palloix et al. 1989
 Nono-Womdim et al. 1991</p> <p>Provvidenti and Gonsalves 1989</p> <p>Singh and Cheema 1989</p> <p>Singh and Singh 1989</p> <p>Narikawa et al. 1989</p> |
|---|---|

Cucumber Mosaic Virus (CMV)

References

41. *C. annuum*. Bangla Green (BG-1), CH-1,
Indonesian selection, Laichi 1, Laichi 2, Lorai,
LS-I, MF41-1, MS-13, Pant C-1, Perennial,
Punjab Lal, S 20-1, Surjamani, Tiwari
— field resistant to CMV
field resistant to leaf curl virus
42. *C. baccatum* LS1621/1, LS1621/2, LS1631/1

C. annuum VC17a, VC40a, VC41a, Lorai, Mexico,
Perennial, Punjab Lal, Serrano Tampiqueno, Szechuan
— also resistant to CVMV
43. *C. frutescens* Pusa Sala Bahar
— also resistant to TMV and leaf curl virus
44. *C. annuum* Vania
— resistant to CMV migration
45. *C. annuum*
AVRDC Acc. C 00266, PBC 199,
Extra Long Selection, Punjab Lal
46. *C. annuum* IHR 243 (Attapadi Lokal), IHR 328-9
(Anekal Lokal), IHR 384 (Perennial), IHR 990
(LICB), IHR 1049 (Puri Red), Pant C-1
resistance stable at temperatures from 26.1
to 37.6°C
also resistant to PVY

C. chinense IHR 1252 (EC 218682)
C. frutescens IHR 1274 (EC 218704)
— also resistant to PVY
47. *C. frutescens*
(several landraces from Latin America)
— involvement of complementary recessive genes
48. *C. annuum* Perennial
resistant to 3 California isolates V 27, causing
severe mosaic and stunting, and V 226 and 28,
causing mild symptoms
susceptible to CMV-FNY, a severe isolate from
New York

Singh and Kaur 1990

Chew and Ong 1990

Tewari 1991

Nono-Womdim et al. 1991

AVRDC 1991

George et al. 1992

Loaiza-Figueros et al. 1992

Lackney et al. 1992

Cucumber Mosaic Virus (CMV)

References

- C. frutescens* BG 2814 (14-6), BG 2816 (16-1)
resistant to mild isolates from California (V 26, V 28) and New York (CMV-Ca, CMV-NY) but not to the FNY isolate
49. *C. annlun.* Indonesian selection, MS-13, Perennial, Punjab Lal
resistant (symptomless) to CMV by artificial inoculation; no local lesions produced after back inoculation to *Clitropodium arnaranticolor*
also resistant under field conditions
- C. annular* 851201, CA 586, ELS-1, BLS-2, Jawahar-218, JCS-1, KCI-152, Laichi 4-4, MF 41-1-2, Ornamental Berry, Ornamental Long, Pant C-1, Surjamani, TC-2
— moderately resistant to CMV (disease incidence ranging from 4 to 24%)
50. *C. annuum* S 20-1, Laichi 2
— also resistant to PVY, TMV, and leaf curl virus
- C. annuurl.* Tiwari
also resistant to TMV and leaf curl virus
also resistant to TMV, PVY, PVMV, and leaf curl virus
- C. annuuar* Lorai
— also resistant to TMV, PVY, leaf curl virus, and CVMV
- C. annuum.* Perennial
— also resistant to TMV, PVY, PVMV, and leaf curl virus
- C. annuum* Punjab Lal
— also resistant to TMV, CVMV, and leaf curl virus
- C. annular* Bangla Green (BG-1)
— also resistant to TMV and leaf curl virus
— partially resistant to PVY
- C. annuum* Ch-1, LS III
— also moderately resistant to leaf curl virus
51. *C. anniunn* TAM Veracruz
(derived from *C. annuum*, AC 2207, Avelar, PI 264281, PI 342947, Jalapeno 1158, Jalapeno100, and Jalapeno L.)
— also resistant to PeMV, PVY, TMV, and TRSV
- Bansal et al. 1992
- Singh 1992
- Villalon et al. 1992

Cucumber Mosaic Virus (CMV)

References

-
52. *C. annuum* HDA 210 bis
 resistant to an isolate from Thailand
 also resistant to CVMV isolates from
 Malaysia and Taiwan
- C. annuum*. I-IDA 273
 resistant to CMV isolates from Taiwan,
 Malaysia, and Thailand
 also resistant to CVMV isolates from Taiwan
 and Malaysia
- C. annuum* HDA 295
 — resistant to an isolate of CMV from Malaysia
 - also resistant to CVMV from Malaysia, Taiwan,
 and Thailand
- C. annuum* Perennial
 — resistant to a CMV isolate from Malaysia
 - also resistant to CVMV isolates from
 Malaysia and Taiwan
- C. annuum* Perennial HDV
 resistant to CMV isolates from Malaysia and Thailand
 also resistant to CVMV isolates from Malaysia,
 Taiwan, and Thailand
53. *C. annuum* Serrano Huasteco
 - also resistant to CVMV

AVRDC 1993

Chew 1993

Curly Top Virus

References

1. <i>C. annuuni</i> PI 257053, PI 281297, PI 288938, PI 357522 apparently resistant when exposed to infectious leafhoppers	Ungs et al. 1977
2. A hybrid from the cross Jerusalem x Sweet Bell resistance controlled by two recessive genes	Konai and Nariani 1980

Leaf Curl Virus Complex'

References

- | | |
|---|---|
| 1. <i>C. annuwn</i> Puri Red, Puri Orange | Mishra et al. 1963
Dhanraj and Seth 1968
Singh 1973 |
| 2. <i>C. annuum</i> EC 4020, EC 6589, EC 7299, ED 7338,
EC 9293, Puri Red, Puri Orange
— field resistant to leaf curl virus | Singh 1973 |
| <i>C. annum</i> Jwala | Tewari and Ramanujam 1974 |
| 4. <i>C. annuum</i> . Surjamani, Perennial
— also resistant to chilli mosaic | Sooch et al. 1976 |
| <i>C. annuum</i> . S 118, S 114
(derived from Perennial x Long Red) | |
| 5. <i>C. annuum</i> Jwala
(derived from Puri Red and N.P. 46A)
— also resistant to mosaic | Tewari 1977, 1982 |
| 6. <i>C. annuum</i> Perennial, S 5-4, S 20-1, S 41-1, S 118-2
- also resistant to TMV and CMV
- 3 weakly linked recessive genes control
resistance | Singh and Thakur 1977 |
| 7. <i>C. annuum</i> Perennial | Singh and Brar 1978
Singh 1992 |
| 8. <i>C. annuum</i> Bangla Green, HS-1, Jamni, Laichi, Longi,
Lorai, Pant C-1, Perennial, S 20-1, S 118-2, Tiwari
— field resistant to leaf curl virus
- also resistant to TMV | Singh and Thakur 1979 |

'The leaf curl virus complex refers to nonmechanically but whitefly-transmitted geminiviruses that usually cause yellowing, leaf curling, and shorting of plants. The viruses causing these symptoms are referred to as leaf curl virus (LCV), tobacco leaf curl virus (TLCV), and chili leaf curl virus in the literature reviewed. Most of these have not been exactly characterized. Resistance listed under 23 refers to tomato yellow leaf curl virus (TYLCV), a well characterized geminivirus which affects mainly tomatoes and only rarely peppers in the Near East. For more detailed information on leaf curl and yellowing viruses of peppers, the reader is referred to the AVRDC Technical Bulletin No. 21: Leaf curl and yellowing viruses of pepper and tomato: an overview.

Leaf Curl Virus Complex

References

9. Unidentified local Indian variety
 tolerant to leaf curl virus
 also tolerant to TMV
 immune to CMV and PVX

 C. annuum Pant C-1, Pant C-2
 — tolerant to leaf curl virus
 - also tolerant to CMV, TMV, and PVX

 C. annuum var. *angulosum*.
 — tolerant to leaf curl virus
 - also tolerant to CMV

10. *C. annuum* Pusa Jwala
 also resistant to TMV
 susceptible to CMV

11. *C. annuum* Delhi Local
 tolerant to leaf curl virus
 also tolerant to TMV
 immune to CMV and PVX

12. *C. annuum*. Cross 218, EC 121490, IC 18253,
 IC 18885, JCA 196, Karanja, Pant C-1
 — less than 30% leaf curl incidence in the field

13. *C. annuum* CA-960, G-4, Jwala

14. *C. annuum* Lorai, Longi, Pant C-1, Perennial, S 118-2
 - resistant/tolerant to leaf curl virus
 - also resistant/tolerant to CMV and TMV

 C. annuum H-6, Longi, No. 76, No. 289-1-3-1-1,
 S 20-1, S 41-1-5, X-200
 — field resistant/tolerant to leaf curl virus
 (disease severity mild to moderate)

15. *C. annuum* Lorai, Longi, Pant C-1, Perennial, S 118-2

16. *C. annuum* S 38-3-19, S 42-2-4
 — tolerant to leaf curl virus
 — also tolerant to PVX and CMV

 C. annuum S 52-1-6, S 96-4-8
 — tolerant to leaf curl virus
 - also tolerant to CMV

Konai and Nariani 1980

Tewari 1982
Singh and Singh 1989Konai and Nariani 1983
Tewari and Viswanath 1986
Tewari 1986

Bhalla et al. 1983

Dhanju 1983

Sharma et al. 1983
Sharma and Singh 1985
Singh 1992

Sharma and Singh 1985

Tewari and Viswanath 1986

Leaf Curl Virus Complex

References

- C. annuum* S 38-2-1, S 38-3-19, S 42-2-4, S 52-1-6, S 96-4-8,
S 96-4-9-3, S 101-2-33
— tolerant to leaf curl virus
- also tolerant to CMV, TMV, and PVX
- C. annuum*. S 81-1-1, S 96-4-9
(selected from Pusa Jwala x Delhi Local)
— tolerant to leaf curl virus
- also tolerant to PVX
17. *C. annuum* Se152-18
(derived from a cross of Pusa Jwala and Delhi Local)
— tolerant to leaf curl virus
also tolerant to CMV and PVX
- C. annuum* Puri Red
- C. annuum* Pusa Jwala
(derived from the cross N.P.46-A and Puri Red)
18. *C. annuum*. Pant C-1, Lic-45, N-66-A
— moderately resistant (40-46% disease incidence)
to leaf curl virus under field conditions
19. *C. annuum* Bangla Green, HS-1, Janmi, Longi,
Lorai, Pant C-1, Perennial, S 20-1, S 118-2
(released as Laichi, Punjab Lal in 1985), Tiwari
— field resistant to leaf curl virus
also resistant to CMV and TMV
20. *C. annuum* Ci 1, Lic-45, N-146
21. *C. annuum* Laichi-2
highly resistant to leaf curl virus
22. *C. annuum* JCA 196, JCA 218, JCA 248, NP-46-A,
Pant C-1, Pusa Jwala
23. *C. annuum* HDA 201, HDA 210 bis, HDA 230, HDA 248,
HDA 249, HDA 252, HDA 260, HDA 268, HDA 273, HDA 295
all lines have *C. annuum* Perennial as a parent
also tolerant to CMV under field conditions
also resistant to PVY (strains 1 and 1.2)
24. *C. annuum*. Jawahar 218
— tolerant to leaf curl virus
- Tewari 1986
- Sawant et al. 1986
- Singh and Kaur 1986
Singh 1992
- Memane et al. 1987
- Singh and Kaur 1987
- Sangar et al. 1988
- Pochard 1988
(personal communication)
- Singh and Cheema 1989

Leaf Curl Virus Complex

References

- C. annuui* n. Punjab Lal
 — resistant to leaf curl virus
 — also resistant to TMV, CMV, and PVY
25. *C. autumn* Bangla Green (BG-1), CI-I-1, Indonesian Selection, Laichi-1, Laichi-2, Lorai, LS-I, MF41-1, MS-13, Pant C-1, Perennial, Punjab Lal, S 20-1, Surjamani, Tiwari-1
 — field resistant to leaf curl virus
 — also field resistant to CMV
26. *C. frutescens* Pusa Sada Bahar
 — also resistant to CMV and TMV
27. *C. annuum* HS III, LS I, LS IV
 Hunda 1992
 (personal communication)
28. *C. annuum* Lorai
 — also resistant to TMV, CMV, PVY, and CVMV
- C. annuum* S20-1, Laichi 2
 — also resistant to TMV, CMV, and PVY
- C. annuum* Tiwari
 — also resistant to TMV and CMV
- C. annuum* Perennial
 — also resistant to TMV, CMV, PVY, and PVMV
- C. annuum* Punjab Lal
 — also resistant to TMV, CMV, and CVMV
- C. annuum* Bangla Green (BG-1)
 also resistant to TMV and CMV
 also partially resistant to PVY
- C. annuum* Indonesian Selection
 — also resistant to CMV and PVY
- C. annuum* CH-1, LS III
 — moderately resistant to leaf curl virus
 — also resistant to CMV
- C. annuum* LS III
 — moderately resistant to leaf curl virus

Pepper Mottle Virus (PeMV)

References

- | | |
|--|--|
| <ol style="list-style-type: none"> 1. <i>C. frutescens</i> Greenleaf Tabasco
(derived from <i>C. frutescens</i> x <i>C. cliinense</i>)
— also resistant to TEV 2. <i>C. annum</i> PI 159236
resistant to PeMV
(multiplication and movement of virus
prevented in homozygous condition)
also resistant to TMV 3. <i>C. annum</i> Delray Bell
— tolerant to PeMV
(resistance controlled by a single recessive gene)
- also tolerant to TEV and PVY 4. <i>C. annum</i> Avelar
— tolerant to PeMV (single recessive gene)
- also resistant to TEV and PVY 5. <i>C. annum</i> Florida VR-4
— also resistant to PVY and TMV 6. <i>C. annum</i> FLBG-1
(derived from Cubanelle and Agronorico-8)
— tolerant to PeMV
- also resistant to TEV and PVY <i>C. annum</i> Florida VR-4
— also resistant to TEV, PVY, and TMV 7. <i>C. airman</i> Tam Mild Jalapeno-1
(derived from <i>C. annum</i> PI 342947, Avelar, and
Jalapeno L)
— also resistant to TEV, TMV, and PVY 8. <i>C. annum</i> Tambel-2, Tarn Mild Chile-2
— also resistant to TEV, PVY, and TMV 9. <i>C. ann.uui</i> PI 152225, PI 159236, Greenleaf Tabasco 10. <i>C. annum</i> Tambel-2
— also resistant to TEV, TMV, and PVY | <p>Zitter 1972
Greenleaf 1986</p> <p>Zitter 1972
Boukema 1977
Subramanya 1982
Greenleaf 1986</p> <p>Zitter and Cook 1973
Cook et al. 1977
Cook 1984a</p> <p>Zitter and Cook 1973
Zitter and Ozaki 1973</p> <p>Cook 1982, 1984a
Abdalla and Desjardins 1985</p> <p>Subramanya et al. 1983</p> <p>Villalon 1983</p> <p>Abdalla and Desjardins 1985</p> <p>Greenleaf 1986</p> <p>Villalon 1986a</p> |
|--|--|

Pepper Mottle Virus (PeMV)

References

11. *C. annum* Tarn Mild Chile-2 (TMC-2)
(derived from *C. anon um* Agronomico-8, Avelar
[both resistant to TEV, PVY, TMV], Cal-Compac 528,
and Anaheim TMR-23)
— also resistant to PVY, TEV, and TMV
12. *C.chinense*
13. *C. alumni* Hidalgo
— also resistant to TEV, TMV, and PVY
14. *C. unman* Hidalgo, Tambel-1, Tambel-2,
Tarn Mild Chile-1, Tam Mild Chile-2,
Tarn Mild Jalapeno-1
— also resistant to TEV, TMV, and PVY
15. *C. annunni* Tarn Rio Grande Gold-Sweet
also resistant to TMV, PVY, and TEV
16. *C. annumn* Criollo de Morelos (CM) 334
— also resistant to PVY
17. *C. antrum* Tam Veracruz
(derived from *C. annum*, AC 2207, Avelar, PI 264281,
PI 342947, Jalapeno 1158, Jalapeno 100, and Jalapeno L.)
— also resistant to CMV, PVY, TMV, TEV, and TRSV

Villalon et al. 1986

Horvath 1986b

Villalon 1986b

Villalon 1986c

Villalbn 1987

Palloix et al. 1991
Chaine Dogiment 1993
Palloix 1993
(personal communication)

Villalon et al. 1992

Pepper Veinal Mottle Virus (PVMV)	References
1. <i>C. annuum</i> DH 2-4, DH 11-15	Rao et al. 1980
2. <i>C. annuum</i> HDA 818, I-IDA 832 (androgenetic doubled haploids resulting from the cross Perennial x Florida VR-2 (both susceptible parents) — resistance to PVMV is a case of complementary gene action	Gebre Selassie et al. 1986
3. <i>C. annuum</i> Perennial	AVRDC 1988
4. <i>C. annuum</i> HDA 832, Perennial HDV	AVRDC 1990a
5. <i>C. annuum</i> HDA 248	AVRDC 1990b
6. <i>C. annuum</i> HDA 248, HDA 249 <i>C. frutescens</i> PSP-11 — also resistant to CVMV <i>C. annuum</i> Perennial HDV, HDA 832 — also resistant to CVMV and PVY	AVRDC 1991
7. <i>C. annuum</i> Perennial — also resistant to TMV, CMV, PVY, and leaf curl virus	Singh 1992

Pepper Vein Banding Virus (PVBV)References

1. *C. annuum*. DH 2-#4, DH 1145
— also resistant to TRSV and PVMV

Rao et al. 1980

Potato Virus X (PVX)

References

-
- | | |
|--|-------------------------------|
| <p>1. Unidentified Indian local variety
 immune to PVX
 also immune to CMV
 - also tolerant to TMV and TLCV</p> <p><i>C. annuum</i>. Pant C-1, Pant C-2
 — tolerant to PVX
 - also tolerant to TMV, CMV, and TLCV</p> <p><i>C. annuum</i> NP 46-A
 <i>C. frutescens</i> EC 31352
 — tolerant to PVX
 - also tolerant to CMV</p> | <p>Konai and Nariani 1980</p> |
| <p>2. <i>C. frutescens</i> 75208
 <i>C. praetermissum</i>. EC 108633
 — high degree of resistance; only necrotic
 local lesions on inoculated leaves, without
 systemic infection</p> <p><i>C. angulosum</i> EC 97758
 <i>C. pendulum</i>
 <i>C. microcarpum</i>
 high degree of resistance to PVX (only necrotic
 local lesions on inoculated leaves, without
 systemic infection)
 also resistant to PVY</p> <p><i>C. annuum</i> var. Kalyan-Sel.
 <i>C. frutescens</i> 76231
 — moderately resistant to PVX
 (no visible symptoms, but virus recovered)</p> | <p>Singh and Chenulu 1980</p> |
| <p>3. <i>C. annuum</i> Pant C-1, Pant C-2
 - tolerant to PVX, CMV, TMV, and leaf curl virus</p> <p><i>C. annuum</i> NP 46-0
 — tolerant to PVX
 - also tolerant to CMV</p> | <p>Konai and Nariani 1983</p> |

Potato Virus X (PVX)

References

-
- | | |
|--|---|
| <p>4. <i>C. (mount</i> Delhi Local
 immune to PVX
 also immune to CMV
 tolerant to leaf curl virus and TMV</p> <p>5. <i>C. annual!</i>. S 38-2-1, S 94-4-9-3, S 101-2-33
 — tolerant to PVX
 - also tolerant to CMV, leaf curl virus, and TMV</p> <p><i>C. annuunt</i> S 38-3-19, S 42-2-4
 — tolerant to PVX
 - also tolerant to CMV and leaf curl virus</p> <p><i>C. aunuuutu</i> S 81-1-1, S 96-4-9
 — tolerant to PVX
 - also tolerant to leaf curl virus</p> <p>6. <i>C. annum</i> Set. 52-18
 (dерived from a cross of Pusa Jwala and Delhi Local)
 — tolerant to PVX
 - also tolerant to leaf curl virus and CMV</p> <p>7. <i>C. angulosunt</i> EC97758
 <i>C. praetennissunt</i> EC108633
 <i>C. frutescens</i> 76 * 208
 — resistance governed by a pair of dominant
 genes</p> | <p>Konai and Nariani 1983
 Tewari and Viswanath 1986
 Tewari 1986</p> <p>Tewari and Viswanath 1986</p> <p></p> <p></p> <p></p> <p></p> <p>Tewari 1986</p> <p>Singh and Chenulu 1984</p> |
|--|---|

Potato Virus Y (PVY)

References

1. *C. amcuuna* P 11 (P1264281)
 unaffected by PVY (several strains)
 reacts in hypersensitive manner to most strains of TMV
 tolerant to most strains of TEV
 Cook and Anderson 1959
 Nagai 1983
2. *C. annuum* Florida VR-2
 (derived from PI 163192, PI 264281, and Cayenne SC 46252)
 also resistant to TEV
et^d gene for resistance (homozygous) confers resistance to mild and severe strains of PVY and to mild strains of TEV
 also resistant to TMV
 homozygous for *L'*
 (determines imperfect localization response to infection with common TMV)
 Cook 1960, 1963b, 1977
 Cook et al. 1976
 Greenleaf 1986
 Sooch et al. 1976
3. *C. annuum* PI 264281
 resistant to PVY-N, PVY-NY⁺, but not to PVY-NY⁻
 which occurs in Puerto Rico
 resistance derived from this introduction is inherited as a single recessive factor; 3 loci associated with resistance.
 Inheritance is monofactorial with the respective dominant alleles conferring susceptibility and the (homozygous) recessive alleles conferring resistance
 also resistant to TEV and TMV
 Cook 1960, 1963b
 IVT 1984
 Nagai 1984
4. *C. annuum* IHR-1049 (Puri Red)
 Anand et al. 1961
5. *C. annuum* Italian EL
 Sinunons 1966
6. *C. annuum* Yolo Y
 (derived from a single mutant plant of Yolo Wonder)
 a single recessive gene *y*^a confers resistance to mild strains of PVY
 also carries *L'* gene for TMV resistance (imperfect localization response to infection with common TMV)
 Cook 1966
 Morales et al. 1976
 Greenleaf 1986
 Pochard 1977

Potato Virus Y (PVY)	References
7. <i>C. anmaem</i> Casca Dura, Mogi-das-Cruzes — immune to some PVY strains tested in Brazil	Nagai 1968
8. <i>C. pubescens</i> 1-30771, 1-30772, SA 112, <i>C. annuum</i> P 11 (PI 264281) — genes y'y' for resistance — immune to PVY strains tested in Brazil	Nagai 1968
9. <i>C. amnani</i> PI 264281 (P11), Agronomico-8, Acc. 2120	Nagai and Smith 1968 Pochard 1987
10. <i>C. annum</i> Agronomico-4, Agronomico-7, Agronomico-8, Agronomico-9, Agronomico-10, Agronomico-11, and Agronomico-12 — resistant to most strains of PVY	Nagai 1968, 1971, 1973, 1980, 1983, 1984 Nagai and Costa 1972
11. <i>C. annum</i> P 11 (PI 264281) — inured to all PVY strains tested in Brazil	Nagai 1968, 1984
12. <i>C. annum</i> PI 264281, Agronomico-10, and Agronomico 11	Nagai 1971
13. <i>C. annum</i> Agronomico-9 and Agronomico-10 (obtained by multiple crossing between <i>C. annuum</i> Casca Dura, Mogi-das-Cruzes, Puerto Rico Wonder)	Nagai 1971, 1980
14. <i>C. baccatum</i> var. <i>pendulum</i> 3-4 — also resistant to CMV	Pochard and Chambonnet 1972 Pochard 1977b Palloix et al. 1988
15. <i>C. annum</i> Agronomico-4 — also resistant to TMV	Nagai 1973
16. <i>C. annum</i> Serrano, Gigante — resistant to all PVY strains tested in Brazil	Pahlen and Nagai 1973
17. <i>C. annum</i> Avelar — tolerance due to a single recessive gene — also resistant to TEV — also tolerant to PeMV	Zitter and Cook 1973 Zitter and Ozaki 1973
18. <i>C. annum</i> . Delray Bell — tolerant to PVY — also tolerant to TEV and PeMV	Zitter and Cook 1973 Cook et al. 1977 Cook 1984a

Potato Virus Y (PVY)	References
19. <i>C. annuum</i> Italian EL, Agronomico, Mogi-das-Cruzes	Lovisolo and Conti 1976
20. <i>C. annuum</i> cv 136 — also resistant to TMV and TEV	Morales et al. 1976
21. <i>C. annuum</i> Florida VR-2 — yy gene for resistance <i>C. baccatum</i> var. <i>pendulum</i> P-7 — virus titer in F ₁ is intermediate between that of the resistant and susceptible parents	Sooch et al. 1976
22. <i>C. annuum</i> Agronomico-8, Avelar, Doux d'Alger, Ikeda, Jalapeno, LP 1, L 136, PI 264281, P 11, Puerto Rico Wonder, Serrano Vera Cruz <i>C. baccatum</i> var. <i>pendulum</i> 3-4 recessive gene for resistance (allelic to Yolo Y resistance) probably there is a single locus with many different alleles <i>C. annuum</i> Rama, Turrialba — hypersensitive reaction (local lesions followed by generalized necrosis) with certain strains of PVY	Pochard 1977a
23. <i>C. annuum</i> Delray Bell also resistant to TEV (<i>et'</i> resistance gene for TEV) also tolerant to PeMV (<i>et''</i> resistance gene for PeMV) develops mild mottle symptoms on leaves, but no fruit symptoms	Cook et al. 1977 Abdalla and Desjardins 1985
24. DH 16-4, DH 30-4	Rao et al. 1980
25. <i>C. annuum</i> Pant C-1 <i>C. angulosum</i> EC 97758	Konai et Nariani 1980
26. <i>C. annuum</i> Chanchal, CM 511, CM 517, CM 526, CM 550, EC 31349, G 2, G 3, G 4, G 5, IC 2345, IC 10435, IC 18205, KA 22, K. No. 2, NP 23, NP 37, NP 42, X 196, X 197, X 198 <i>C. frutescens</i> 76 * 55, 76 = 252	Singh and Chenulu 1980

Potato Virus Y (PVY)

References

- C. angulosum* EC 97758
C. microcarpum.
C. pendulum
 — resistant to PVY
- C. annuum* NP 36, IC 3430
C. frutescens 76 * 201, 76 * 208
C. pubescens
 — moderately resistant to PVY
27. *C. annuum* Florida XVR 3-25, USAJI-5
 — also resistant to TEV
 Cook 1982
28. *C. annuum*. Florida XVR 3-25
 (derived from *C. chacoense* PI 260435 x *C. annuum* VR-2)
 — carries *et*^a and *L'* resistance genes
 - also resistant to TMV
 Cook 1982, 1984c
29. *C. annuum* Florida VR-4
 (derived from VR-2 x Delray Bell)
 also resistant to TEV and PeMV
 susceptible to common TMV
 carries resistance genes *et'* and *et*^a
 Cook 1982, 1984a
30. *C. annuum* Todo Ano
 Ferreira et al. 1982
31. *C. annuum* Serrano Vera Cruz
 - resistant to PVY strains 0, 1, 1.2
 Gebre Selassie et al. 1983
32. *C. annuum* FLBG-1
 (derived from Cubanelle and Agronomico-8)
 — also resistant to TEV
 - also tolerant to PeMV
 Subramanya et al. 1983
- C. annuum* Florida VR-4
 — also resistant to TEV and TMV
 Subramanya et al. 1983
33. *C. annuum* Agronomico-10G, Agronomico-13,
 Agronomico-14, Agronomico-15,
 Agronomico-16, Magda, Margareth
 Nagai 1983
34. *C. annuum*. Tam Mild Jalapefno-1
 [derived from *C. annuum* PI 342947 (resistant to
 TEV, PVY, and TMV), Avelar (resistant to TEV
 and PVY), and Jalapeno I]
 — also resistant to TEV, TMV, and PeMV
 Villalon 1983

Potato Virus Y (PVY)	References
35. <i>C. nungu</i> t Florida VR 2-34 also resistant to TEV and TMV carries resistance genes <i>el'</i> and <i>L'</i>	Cook 1984b Greenleaf 1986
36. <i>C. annuum</i> USAJI 5 (derived from PI 265281, resistant to TEV and PVY [<i>ell</i> and a selection from ECUAJI, resistant to PVY) — also resistant to TEV	Cook 1984d Greenleaf 1986
37. <i>C. annuum</i> ECUAJI (from Ecuador)	Cook 1984d
38. <i>C. baccatum</i> var. <i>pendulum</i> <i>C. chinense</i> <i>C. eximium</i> <i>C. flexuosum</i> — resistant to PVY by hypersensitive type of reaction	Horvath 1984
39. <i>C. annuum</i> Tambel-2, Tam Mild Chile-2 (TMC-2) — also resistant to TEV, PeMV, and TMV	Abdalla and Desjardins 1985
40. <i>C. annuum</i> Tam Mild Chile-2 (TMC-2) (derived from Agronomico-8 and Avelar [both resistant to TEV, PVY, and TMV], Cal-Compack 528 and Anaheim TMR-23) — also resistant to PeMV, TMV, and TEV	Villalon et al. 1986a
41. <i>C. annuum</i> Florida VR-2, Lamuyo, Early Prolific, California Wonder 300	Apablaza and Jorquera 1985
42. <i>C. annuum</i> var. <i>angulosum</i> EC 97758 <i>C. baccatum</i> var. <i>nticropinn</i> — single pair of identical recessive genes <i>C. annuum</i> cv. NP 36 <i>C. frutescens</i> 76 = 208 <i>C. pubescens</i> moderately resistant (monogenic recessive)	Singh and Chenulu 1985
43. <i>C. annuum</i> Long Green Chile 76042, Tam Mild Jalapeno 1, Texas Bell 17004, — also resistant to TEV, PeMV, and TMV	Greenleaf 1986
44. <i>C. annuum</i> Agronomico-8, Agronomico-10, Avelar, Casca Dura, Ikeda, Moura, PI 264281 (P11), Puerto Rico Wonder	Greenleaf 1986

Potato Virus Y (PVY)

References

-
- | | |
|---|----------------------------------|
| <p>45. <i>C. cltinense</i>
 <i>C. baccatum</i> var. <i>pendulum</i>
 <i>C. eximium</i>
 <i>C. flexuosum</i>
 — all show local lesions and leaf drop</p> <p><i>C. pubescens</i>
 — shows local lesions and leaf drop
 - also resistant to ToMV and CMV</p> | <p>Horvath 1986a</p> |
| <p>46. <i>C. cardenasii</i>
 <i>C. frutescens</i></p> | <p>Horvath 1986b</p> |
| <p>47. <i>C. annuum</i> S 38-2-1, S 38-3-19, S 42-2-4,
 S 52-1-6, S 96-4-9-3, S 96-4-8, S 101-2-33
 — tolerant to PVX
 — also tolerant to TMV, CMV, and TLCV</p> <p><i>C. annuum</i> S 81-1-1, S 96-4-9
 — also tolerant to PVX and leaf curl virus</p> | <p>Tewari and Viswanath 1986</p> |
| <p>48. <i>C. annuum</i> Tambel-2
 also resistant to TMV, TEV, and PeMV</p> | <p>Villalon 1986a</p> |
| <p>49. <i>C. annuum</i> Tarn Mild Chile-2 (TMC-2)
 (derived from Agronomico-8 and Avelar [both
 resistant to TEV, PVY, and TMV], Cal-Compact 528
 and Anaheim TMR 23)
 — also resistant to TEV, TMV, and PeMV</p> | <p>Villalon et al. 1986</p> |
| <p>50. <i>C. annuum</i> Hidalgo
 (derived from PI 342947 [resistant to TEV, PVY,
 and TMV] x Avelar [resistant to TEV and PVY])
 — also resistant to TEV, TMV, and PeMV</p> | <p>Villalon 1986b</p> |
| <p>51. <i>C. annuum</i>. Hidalgo, Tambel-1, Tambel-2,
 Tam Mild Chile-1, Tam Mild Chile-2,
 Tam Mild Jalapeno 1
 — also resistant to TEV, TMV, and PeMV</p> | <p>Villalon 1986c</p> |
| <p>52. <i>C. annuum</i> Tarn Rio Grande Gold-Sweet
 - also resistant to TEV, TMV, and PeMV</p> | <p>Villalon 1987</p> |

Potato Virus Y (PVY)

References

-
53. *C. mum*' PM 4
(F hybrid, Northrup Seed Co.)
— single dominant gene for resistance to all PVY strains tested in Brazil, including a new strain,
54. *C. annuum* Redlands Sweet Sue
(derived from Yolo Y x Hungarian Yellow Wax)
- resistant to Australian strain 1 (but not to strains 2 or 3)
55. *C. annuum* Perennial
partly resistant (highly tolerant) to several PVY strains
also resistant to virus strain EP 11, which is serologically distinct from other PVY strains and virulent on most of the resistant lines
also tolerant to CMV under field conditions
- C. annuum* HDA 201, HDA 210bis, HDA 230, HDA 248, HDA 249, HDA 252, HDA 260, HDA 268, HDA 273, HDA 295
(all lines have Perennial as a parent)
resistant to PVY strains 0, 1, and 1.2
also tolerant to CMV and tomato yellow leaf curl virus (TYLCV) under field conditions
56. *C. annuum* YoloY, Maor, Redlands Sweet Sue
- resistant to PVY strain 1 in Australia
- C. annuuan* Bell Tower
— resistant to PVY strain 1 in Australia
- partially resistant to PVY strain 2 in Australia
- C. annuum* Delray Bell, Florida VR-4
— resistant to PVY strain 1, 2, and 3 in Australia
- C. trnnuum* Ace. 1534 (piquin type chilli)
— highly resistant to PVY strains 1, 2, and 3 in Australia
57. *C. annuum* Agronomico 10G, Magda, Margareth
58. *C. annuum* Punjab Lal
— also resistant to TMV, CMV, and leaf curl virus

Potato Virus Y (PVY)

References

-
- | | |
|--|--|
| <p>59. <i>C. annuum</i> Hidalgo, Rio Grande Gold
 <i>C. baccatum</i> PEN 3-1
 Unknown species AVRDC Acc. C 00006, C 00574</p> <p><i>C. ammm.</i> HDA 273
 — also resistant to CVMV</p> <p><i>C. annuum</i> HDA 832, Perennial HDV
 — also resistant to CVMV and PVMV</p> | <p>AVRDC 1991</p> |
| <p>60. <i>C. autumn/</i> Criollo de Morelos (CM) 334
 - rapid blockage of virus infection, associated
 with local necrotic lesions
 dominant gene (Cy-2) controls resistance to
 pathotypes PVY 0, 1, and 1-2
 also resistant to PeMV
 also carries cy-1 gene that confers recessive
 resistance to PVY</p> | <p>Palloix et al. 1991
 Palloix 1993
 (personal communication)
 Chaîne-Dogiment 1993</p> |
| <p>61. <i>C. annuum</i> AVRDC Acc. C 01175
 (Criollo de Morelos 334)
 - also resistant to CVMV</p> | <p>Green 1991
 (unpublished)</p> |
| <p>62. <i>C. annuum</i> AVRDC Acc. C 00229-1, C 00753,
 C 01511, C 05116; Thailand lines Kee Noo Chinda
 No. 1, Chinda No. 23, Chinda No. 24, Hua Rua No. 1,
 Hua Rua No. 2, Hua Rua No. 3, Sri Saket
 Unknown species AVRDC Acc. C 00352, C 00373,
 C 00574, C 00595, C 01176, C 05078</p> <p><i>C. chinense</i> AVRDC Acc. C 00304
 — also resistant to ToMV, TMV, and PMMV</p> <p><i>C. baccatum</i> var. <i>pendulum</i> AVRDC Acc. C 00803
 Unknown species AVRDC Acc. C 01173
 — also resistant to ToMV</p> <p><i>C. baccatum</i> var. <i>pendulum</i> AVRDC Acc. C 00753-1,
 C 01172
 <i>C. chinense</i> AVRDC Acc. C 00940
 <i>C. frutescens</i> AVRDC Acc. C 00848
 Unknown species AVRDC Acc. C 01090, C 01175
 — also resistant to TMV and ToMV</p> | <p>AVRDC 1992</p> |

Potato Virus Y (PVY)

References

-
63. *C. annuum* Lorai, Perennial, S 20-1
(selection from a cross of Perennial x Long Red)
— also resistant to CMV, TMV, and leaf curl virus
- C. annuum*. Punjab Lal
— partially resistant to PVY
— also resistant to CMV, TMV, and leaf curl virus
- C. annuum* Indonesian Selection
— also resistant to CMV and leaf curl virus
64. *C. annuum* IHR 993 (EC 171502), IHR 994 (IAC-8)
C. frutescens IHR 1243 (EC 218673)
C. pubescens IHR 1267 (EC 218692)
- C. annuum* IHR 243 (Attapadi Lokal),
IHR 328-9 (Anekal Lokal), IHR 384 (Perennial),
IHR 990 (LIC B), IHR 1049 (Puri Red), Pant C-1
C. chinense IHR 1252 (EC 218682)
C. frutescens IHR 1274 (EC 218704)
— also resistant to CMV
65. *C. annuum* Laichi 2, S 20-1
— also resistant to TMV, CMV, and leaf curl virus
- C. annuum* Perennial
— also resistant to TMV, CMV, PVMV, and leaf curl virus
- C. annuum* Lorai
— also resistant to TMV, CMV, CVM, and leaf curl virus
- C. annuum* Indonesian Selection
— also resistant to CMV and leaf curl virus
- C. annuum* Bangla Green (BG-1)
— partially resistant to PVY
— also resistant to TMV, CMV, and leaf curl virus
66. *C. annuum* Yolo Y
resistant to PVY 0 but not to PVY 1
carries gene *vy* for resistance
- C. annuum* Italian EL
resistant to PVY-0, but not PVY-1
resistance is recessive and monogenic
resistance gene(s) the same or allelic to
Yolo Y resistance genes

Potato Virus Y (PVY)

References

- C. none um* Florida VR-2
— resistant to PVY-1 but not PVY 1-2
- C. armuum* Serrano Vera Cruz
— resistant to PVY-0, 1, and 1-2
67. *C. annuuan* Tam Veracruz
(derived from *C. annu.um*, AC 2207, Avelar, PI 264281, PI 342947, Jalapeno 1158, Jalapeno 100, and Jalapeno L.)
— also resistant to CMV, PeMV, TEV, TMV, and TRSV
Villalon et al. 1992
- C. annurrm* Avelar, PI 342947
— also resistant to Texas isolates of TEV and TMV
68. *C. annuunr* AVRDC Acc. C 01175 (Criollo de Morelos 334), C 01276, PBC 392, Serrano Tampiqueno, Thailand Lines Chinda No 1, Hua Rua, Lueng 1, Lueng 2, Lueng 3, Lueng 4, Lueng 7, Lueng 8, Lueng 9, Lueng 10, Tupang
AVRDC 1993
- C. frutescens* AVRDC Acc. C 05459, C 05550, C 05560, C 05570, C 05571, C 05572
- C. annrann* AVRDC Acc. C 05342
C. frutescens AVRDC Acc. C 05473
— also resistant to ToMV
69. *C. annuum* PBC 349, PBC 365, PBC 370, PBC 450, PBC 454, PBC 458, PBC 483, PBC 495, PBC 521, PBC 524, PBC 534, PBC 535, PBC 545, PBC 580, PBC 581, PBC 582, PBC 596, PBC 622, PBC 710, PBC 729, PBC 730, PBC 731, PBC 787, PBC 830
AVRDC 1994
- C. nnnuunt* AVRDC Acc. C 01664
— also resistant to CVMV

Tobacco Etch Virus (TEV)

References

-
- | | |
|---|--|
| 1. <i>C. chinense</i> PI 152223 | Greenleaf 1953 |
| 2. <i>C. annuum</i> . SC 46252
resistant, but not immune to TEV
single recessive gene for resistance | Greenleaf 1953, 1956
Cook 1960 |
| 3. <i>C. chinense</i> PI 152225 (from Peru)
— resistant, but not immune to TEV | Greenleaf 1953, 1959 |
| 4. <i>C. annuum</i> PI 264281
— single recessive gene for resistance
— also resistant to PVY and TMV | Greenleaf 1956
Cook 1960 |
| 5. <i>C. frutescens</i> Tabasco-G (Greenleaf Tabasco)
(derived from <i>C. frutescens</i> Tabasco, <i>C. chinense</i>
PI 152225 and PI 159236)
— monogenic resistance
— also resistant to PeMV | Greenleaf 1956, 1975
Zitter 1972 |
| 6. <i>C. annuum</i> P 11
tolerant to most strains of TEV
unaffected by PVY (several strains)
also resistant to TMV
(hypersensitive resistance to most strains of TMV) | Cook and Anderson 1959 |
| 7. <i>C. annuum</i> Acc. 2120, Agronomico-8
(a derivative of PI 264281 [P 11])

<i>C. frutescens</i> PI 152225, PI 159236 | Nagai and Smith 1968 |
| 8. <i>C. frutescens</i> LP-1
— resistance to TEV dominant
— also resistant to CMV (recessive) | Barrios et al. 1971 |
| 9. <i>C. frutescens</i> Greenleaf Tabasco
— also resistant to PeMV | Zitter 1972
Greenleaf 1986 |
| 10. <i>C. annuum</i> Delray Bell
— also resistant to PeMV and PVY | Zitter and Cook 1973
Cook et al. 1977
Cook 1984a |

Tobacco Etch Virus (TEV)

References

-
- | | |
|--|--|
| <p>11. <i>C. annuum</i> PI 264281 (P 11)
— immune to TEV</p> <p>12. <i>C. annuum</i> cv. 136
— also resistant to TMV and PVY</p> <p>13. <i>C. alumni</i> Avelar
— TEV tolerance due to a single recessive gene
- also resistant to PVY and PeMV</p> <p>14. <i>C. annuum</i> VR-2, Delray Bell
— also resistant to TMV and PVY</p> <p>15. <i>C. annuum</i> Florida VR-2, VR-4, Delray Bell
— also resistant to PVY and PeMV</p> <p>16. <i>C. annuum</i> PI 342947 (Acc. 2120), PI 410407 (Avelar),
PI 432946 (Agronomico-8)
<i>C. chinense</i> PI 152225
— highly resistant to TEV</p> <p><i>C. annuum</i> PI 264281 (P 11), PI 432948
moderately resistant to TEV</p> <p><i>C. unman</i> PI 342949 (Casca Dura)
resistant in the greenhouse but not in the field</p> <p>17. <i>C. annuum</i> Florida XVR 3-25, USAJI-5
— also resistant to PVY</p> <p><i>C. annuum</i> Florida VR-4
— also resistant to PeMV, PVY, and TMV</p> <p>18. <i>C. annuum</i>. FLBG-1
(derived from Cubanelle and Agronomico-8)
— also resistant to PVY
- also tolerant to PeMV</p> <p><i>C. annuum</i> Florida VR 4
— also resistant to TMV and PVY</p> <p>19. <i>C. annuum</i> Tam Mild Jalapeño-1
(derived from <i>C. annuum</i> PI 342947, Avelar, and
Jalapeño L)
— also resistant to TMV, PVY, and PeMV</p> | <p>Weinbaum and Milbrath 1976</p> <p>Morales et al. 1976</p> <p>Zitter and Cook 1973
Zitter and Ozaki 1973</p> <p>Cook 1977</p> <p>Cook et al. 1977
Cook 1984a</p> <p>Sowell and Demski 1977</p> <p>Cook 1982</p> <p>Subramanya et al. 1983</p> <p>Villalon 1983</p> |
|--|--|

Tobacco Etch Virus

References

- C. annuuna* Long Green Chile 76042, Texas Bell
— also resistant to Texas strains of SL-TMV, PVY, and PeMV
20. *C. annuuna*. Florida VR 2-34
— also resistant to PVY and TMV
21. *C. antrum* USAJI 5
(derived from PI 265281, resistant to TEV and PVY [et^a] and a selection from ECUAJI, resistant to PVY)
— also resistant to PVY
22. *C. annuulaa*. California Wonder 300, Early Prolific, Florida VR-2, Lamuyo
resistant to PVY common strain (PVY^o) in Chile
23. *C. annuu a* Tambel-2, Tam Mild Chile-2 (TMC-2)
- also resistant to PVY, PeMV, and TMV
24. *C. chinense* AC 2176 TEV
- also resistant to TMV
25. *C. annular'* Cayenne SC 46252, Pimsan 3
resistant to TEV (et' gene for resistance)
- C. annuuna* Cayenne 16, Cayenne 17, Cayenne 18, Cayenne 19, Cayenne 20
— also resistant to PeMV
26. *C. annuunl* Tam Mild Chile-2 (TMC-2)
(derived from Agronomico 8, Avelar [both resistant to TEV, PVY, and TMV], Cal Compack 528, and Anaheim TMR-23)
— also resistant to PVY, PeMV, and TMV
27. *C. annannnn* Tambel-2
- intermediate resistance
- also resistant to TMV, PVY, and PeMV
- C. alumni* Agronomico-8
moderately resistant
28. *C. annuunl*. Hildalgo, Tambel-1, Tambel-2, Tam Mild Chile-1, Tam Mild Chile-2, Tam Mild Jalapeno-1
— also resistant to PeMV, TMV, and PVY
- Cook 1984b
- Cook 1984d
Greenleaf 1986
- Apablaza and Jorquera 1985
- Abdalla and Desjardins 1985
- Kounovasky et al. 1985
- Greenleaf 1986
- Villalon et al. 1986
- Villalon 1986a
Padgett et al. 1987
- Villalon 1986c
Villalon et al. 1986

Tobacco Etch Virus

References

-
- | | |
|---|--|
| <p>29. <i>C. chinense</i></p> <p>30. <i>C. annuunt</i> Hidalgo
(derived from PI 342947 [resistant to TEV, PVY, and TMV] x Avelar [resistant to TEV and PVY])
— also resistant to PVY, PeMV, and TMV</p> <p>31. <i>C. annutnt</i> Tam Rio Grande Gold-Sweet
— also resistant to PeMV, PVY, and TMV</p> <p>32. <i>C. anruain</i>. Georgia C 44NV, Georgia C 44CA
— highly resistant under greenhouse conditions
but mild mottle in 10% of the plants in the field</p> <p>33. <i>C. chinense</i> PI 159241</p> <p>34. <i>C. annuuni</i> GA-C44-V22
highly resistant under greenhouse conditions
(no symptoms; no virus detected by ELISA)
highly tolerant under field conditions
(mild mottle observed in the field in 15%
of the plants, but virus was not detected)</p> <p><i>C. annuuni</i> Florida XVR-3-25
difficult to infect mechanically
highly resistant under greenhouse conditions
(no symptoms; no virus detected by ELISA)
mild disease in the field (50-85% of plants
developed mild symptoms in the field.
Virus could be detected in less than 15%
of the plants with symptoms)</p> <p><i>C. annutnn</i> Tambel-2, Asgrow-XPB-5021
moderate resistance (low to medium virus
titer for 2-3 weeks after inoculation)</p> | <p>Horvath 1986b</p> <p>Villalon 1986b</p> <p>Villalon 1987</p> <p>Kuhn et al. 1987</p> <p>Padgett et al. 1987</p> <p>Kuhn et al. 1987, 1989</p> |
|---|--|
35. *C. an uturn* Tam Veracruz
also resistant to CMV, PVY, PeMV, TMV, and TRSV
- C. annuunt* Avelar, PI 342947
— also resistant to Texas isolates of PVY and TMV

Tobacco Ring Spot Virus (TRSV)	References
1. <i>C. annuum</i> Tarn Veracruz (derived from <i>C. annuum</i> , AC 2207, Avelar, PI 264281, PI 342947, Jalapeno 1158, Jalapeno 100, and Jalapeno L.) — also resistant to CMV, PeMV, PVY, TEV, and TMV	Villalon et al. 1992
2. <i>C. annuum</i> DH 2-4, DH 11-15 - also resistant to PVBV and PVMV	Rao et al. 1980

Tobamoviruses'

References

-
- | | |
|--|---|
| 1. <i>C. baccatum</i> . PI 152234
<i>C. chinense</i> PI 152222, PI 152225
Unlisted species PI 152221, PI 152235, PI 152453
— local lesions | Greenleaf 1953 |
| 2. <i>C. chinense</i> PI 152225, PI 159236,
PI 159241, PI 257284, PI 315008, PI 315024,
PI 315023 | Greenleaf 1956
Boukema 1977 |
| 3. <i>C. annuum</i> P11 (PI 264281)
reacts in a hypersensitive manner to most strains
of TMV
tolerant to most strains of TEV
unaffected by PVY (several strains) | Cook and Anderson 1959 |
| 4. <i>C. annuum</i> Florida VR-2
— resistant to TMV
— also resistant to strains of PVY and TEV | Cook 1960, 1963b, 1977
Cook et al. 1976
Greenleaf 1986
Sooch et al. 1976 |
| 5. <i>C. annuum</i> PI 264281
— also resistant to PVY and TEV | Cook 1960, 1963b
IVT 1984
Nagai 1984 |
| 6. <i>C. annuum</i> PI 163201
necrotic local lesions on inoculated leaves
no systemic infection | Pound and Singh 1960 |
| 7. <i>C. annuum</i> PI 183441
a plant of this introduction remained
symptomless after repeated inoculations | Sowell 1960
Cook 1963a |
| 8. <i>C. annuum</i> cv. 136
— also resistant to TEV and PVY | Cook 1966
Morales et al. 1976 |
-

'Several tobamoviruses are known to infect peppers (see Appendix 1), the most common of which are tobacco mosaic virus (TMV), tomato mosaic virus (ToMV), and pepper mild mottle virus (PMMV). Some literature references, particularly the early ones, do not distinguish between the different tobamoviruses but refer only to TMV, while they may actually be dealing with ToMV or PMMV. For information on the different strains and pathotypes of the tobamoviruses, the reader is referred to the AVRDC Technical Bulletin No. 18: Characteristics and control of viruses infecting peppers: a literature review.

Tobamoviruses	References
9. <i>C. annuum</i> Yolo Y — also resistant to PVY	Cook 1966 Morales et al. 1976 Greenleaf 1986 Pochard 1977
10. <i>C. frutescens</i> Tabasco	Ramakrishnan et al. 1966
<i>C. Inicrocarpul</i> † — immune to TMV	
<i>C. annuum</i> Birdeye, 971 — hypersensitive reaction with TMV	
11. <i>C. annuum</i> Agronomico-9, Agronomico-10, Agronomico-11, Agronomico-12 - highly resistant to TMV - also resistant to PVY	Nagai and Costa 1972 Nagai 1973
12. <i>C. chinense</i> PI 159236 — also resistant to PeMV	Zitter 1972 Boukema 1977 Subramanya 1982
13. <i>C. annuum</i> Agronomico-4	Nagai 1973
14. <i>C. annuum</i> A ₂₇ , Cl, IC 3425, IC 3440, IC 3588, IC 3779, Konidiverum G2, Puri Orange, Puri Red, Surjamukhi — field resistant to TMV	Singh 1973
16. <i>C. annuum</i> S 5-4, S 20-1, S 41-1, S 41-2, S 41-3, S 69-1, S 72-3, S 72-4	Kaul 1975
17. <i>C. annuum</i> INRA Bird's Eye, Lanulyo, Yolo Wonder, Yolo Y, Toledo F ₁	Lovisolo and Conti 1976
18. <i>C. annuum</i> Yolo Wonder, Yolo Y - genotype L'/L' - also resistant to PVY	Greenleaf 1986 Morales et al. 1976
<i>C. annuum</i> cv. 136 — also resistant to TEV and PVY	
18. <i>C. annunni</i> Fusimi, Kimjang gochu, Sae gochu, Sandia, 2520, Taekuk, Wheonggeri 128-2-1, Wheonggeri 128-3-3, Yolo Y	Kang and Choi 1976

Tobainoviruses

References

- C. frutescens* Tabasco 131-4
 — local lesions only in inoculated leaves
 - possibly carries the *L* gene for resistance
19. *C. chinense* PI 152225
 — 1 dominant gene
 van den Berkmortel 1977
20. *C. alumni* Florida VR-2
 - also resistant to TEV and PVY
 Cook 1977;
 Greenleaf 1986
21. *C. baccatunt*
 resistant to strains P 0 and P 1
 susceptible to strain P 1.2
 Rast 1977
22. *C. annuum* Perennial
 — also resistant to CMV and leaf curl virus
 — 3 weakly recessive genes control resistance to CMV, TMV, and leaf curl virus
 Singh and Thakur 1977
23. *C. annuum* Kalocsai VI
 (derived from a cross between a half wild hypersensitive Chinese variety and the susceptible variety Szeged 91-95)
 — also tolerant to CMV
 Szirmai 1978
24. *C. annuum* Keystone Resistant Giant
 - *L'* gene for resistance
 Zatyko 1978
25. *C. chinense* (10 Pules)
 - L^3 gene for resistance
 Boukema 1978, 1980
- C. chinense* PI 152225, PI 159233, PI 159236,
 PI 213917, PI 215024, PI 224424, PI 257284,
 PI 315008, PI 315023
 — local necrotic lesions with P 8 st'ain
26. *C. annuum* Bangla Green, HS-1, Jamni, Laichi, Longi, Lorai, Pant C-1, Perennial, S 20-1, S 118-2, Tiwari
 — also resistant to CMV
 - also field resistant to leaf curl virus
 Singh and Thakur 1979
27. *C. annuum* Kremovyi 20 (Cream 20), Yolo Wonder, Adygei, Konservnyi, Krasnyi, Red Canning
C. baccatum var. *pendulum*
C. chinense
C. frutescens
 Timina and Balashova 1979

Tobamoviruses

References

28. *C. annuli*,^{'''} Feherozon
— *L'* gene for resistance
Zatyko et al. 1979
Tobias and Csillery 1982
29. *C. chinense* PI 159236, PI 213917, PI 224424,
PI 257117, PI 257284, PI 281428,
PI 315008 (Miscucho), PI 315023 (Mishme),
PI 315024, PI 159233 (listed *C. annuum*),
PI 15362, PI 5380, PI 5393, PI 5395, PI 5397,
PI 5401, PI 5416, PI 5431, PI 5451, PI 5490,
PI 7204, PI 7208, W1.0, D8
genotype L^3L^3
resistant to tomato strain P 0 and "pepper strains"
P 1 and P 1.2, but not to P 1.2.3.
- C. chinense* Ru 72-292
resistant to tomato strain P 0
genotype $L'L'$
- C. frutescens* Tabasco
— L^2L^2 genotype
— resistant to P 0 and P 1
- C. chinense* PI 152225
— segregates 1 susceptible, 8 resistant
- C. chinense* PI 159241
— segregates 2 susceptible, 7 resistant
— 1 dominant gene
30. *C. annuum* Pant C-1, Pant C-2
— also resistant to CMV, PVX, and leaf curl virus
Konai and Nariani 1980
- Unidentified local Indian variety
— tolerant to TMV
— also immune to CMV and PVX
31. *C. annuum* Gypsy (F_1 hybrid)
Tigchelaar 1980
32. *C. chinense* PI 152225, PI 159230, PI 315008,
PI 315023
— resistant by hypersensitivity to a
new strain isolated from *C. annuum* Lamuyo
Gebre Selassie et al. 1981
33. *C. chinense* (7 accessions)
resistant to a strain of TMV; isolated from
C. frutescens Tabasco
Boukema 1982

Tobamoviruses

References

-
- | | |
|---|---|
| <p>34. <i>C. chacoense</i> PI 260429, SA 185
— L" gene for resistance</p> <p>35. <i>C. annuton</i> Florida XVR 3-25
— also resistant to PVY and TEV</p> <p>36. <i>C. annuurn</i> Florida VR 4
— also resistant to PVY, TEV, and PeMV</p> <p>37. <i>C. annuuut</i> Yolo Wonder, Florida VR-2
resistant to pathotype TMV-PO</p> <p>38. <i>C. annuun</i> Keystone
— resistant to a Cuban strain of TMV</p> <p>40. <i>C. chacoense</i> SA 185
resistant to TMV strain P 1.2.3
resistance is based on 1 incompletely
dominant gene</p> <p>41. <i>C. annuurn</i> KKG-W3, Gu nzulina, Poivron, Titan
— resistant to TMV-T</p> <p>43. <i>C. alumni</i> Garden Sunshine
(derived from Yolo Wonder x Aconcaqua)</p> <p>44. <i>C. autumn</i> Pusa Jwala
— also resistant to leaf curl virus and CMV</p> <p>43. <i>C. baccatum</i> PI 260549</p> <p><i>C. chinense</i> PI 159233 (USA), PI 159236 (USA), PI 257284
(Spain), PI 315008 (Miscucho), PI 315024 (Peru)
— resistant to tomato and pepper strains</p> <p><i>C. anniutnt</i> Rama, Turrialba
— resistant to pepper strain P 11</p> <p>58 species
— resistant to TMV-T and TMV-Tm</p> <p>31 species
— resistant to TMV-P 11 (pepper strain)</p> <p>5 species
— resistant to TMV-P 8 (pepper strain)</p> | <p>Boukema 1982, 1984
Tobias and Csillery 1982</p> <p>Cook 1982</p> <p>Cook 1982, 1984</p> <p>Dumas de Vaulx et al. 1982</p> <p>Cordero and Gaborjanyi 1982</p> <p>IVT 1982</p> <p>Kostova 1982</p> <p>Millet and Jones 1982</p> <p>Tewari 1982
Singh and Singh 1989</p> <p>Rast 1982</p> |
|---|---|

Tobamoviruses	References
44. <i>C. annuum</i> S 38-2-1, S 94-4-9-3, S 101-2-33 — tolerant to TMV — also tolerant to CMV, PVX, and leaf curl virus	Tewari and Viswanath 1982
45. <i>C. frutescens</i> 33/5 — dominant oligo gene Line L 938 — single recessive gene	Balashova et al. 1983
46. <i>C. annuum</i> (54 lines) resistant by hypersensitive reaction	Marie 1983
47. <i>C. annuum</i> . Lorai, Longi, Pant C-1, Perennial, S 118-2 resistant/tolerant to TMV also resistant/tolerant to CMV and leaf curl virus	Sharma et al. 1983 Sharma and Singh 1985
48. <i>C. annuum</i> Pant C-1, Pant C-2 — tolerant to TMV — also tolerant to CMV, PVX, and leaf curl virus <i>C. annuum</i> Delhi Local — tolerant to TMV also tolerant to leaf curl virus immune to CMV and PVX	Konai and Nariani 1983
49. <i>C. annuum</i> Tam Mild Jalapeno-1 (derived from <i>C. annuum</i> . PI 342947, Avelar, and Jalapeno L) — also resistant to TEV, PVY, and PeMV	Villalon 1983
50. <i>C. annuum</i> Florida VR 2-34 — also resistant to PVY and TEV	Cook 1984b Greenleaf 1986
51. <i>C. annuum</i> . Florida XVR 3-25 (derived from <i>C. chacoense</i> PI 260435 x <i>C. annuum</i> Florida VR-2) — also resistant to PVY	Cook 1984c
52. <i>C. baccatum</i> var. <i>baccatum</i> . <i>C. baccatum</i> var. <i>pendulum</i> <i>C. frutescens</i> var. <i>minimum</i> . <i>C. frutescens</i> Tabasco <i>C. praetermissum</i> <i>C. pubescens</i>	Horvath 1984

Tobamoviruses

References

-
- | | |
|---|--|
| <p>53. <i>C. chacoense</i> PI 260429
 resistant to TMV strain P1.2.3
 resistance is based on 1 single incompletely
 dominant gene L'</p> <p>54. <i>C. annuum</i> Torito
 — resistant to TMV-0</p> <p>55. <i>C. annuum</i>. L 290, L 305, L 323</p> <p>56. <i>C. annuum</i>. Tam Mild Chile-2 (TMC-2)</p> <p>57. <i>C. annuum</i> Tambel-2
 also resistant to PVY, PeMV, and TEV</p> <p>58. F₃ generation
 (Ahnnapaprika and Paradicsomalaku Zold Szentesi x
 Florida VR-2)</p> <p>59. <i>C. annuum</i> Wongkyo 306</p> <p>60. F₄ generation
 (<i>C. annuum</i> Athena ns8 x <i>C. chinense</i> AC 2176)</p> <p>61. <i>C. chacoense</i></p> <p>62. <i>C. annuum</i> Bighart, Pimsan 1, Pimsan 2
 L gene for TMV resistance</p> <p> <i>C. annuum</i> Florida VR-2, Florida VR 2-34,
 Florida XVR 3-25, Keystone Resistant Giant,
 Verbeterde Glas, Yolo Wonder, Yolo Y
 — L'/L' genotype
 - resistant to strains of TMV (P0)</p> <p> <i>C. frutescens</i> Tabasco</p> | <p>IVT 1984</p> <p>Stolk 1984</p> <p>Jilaveanu et al. 1984</p> <p>Abdalla and Desjardins 1985
 Villalon 1986a, c</p> <p>Abdalla and Desjardins 1985
 Villalon 1986a, c</p> <p>Barta et al. 1985</p> <p>Choi and Pae 1985</p> <p>Kounovasky et al. 1985</p> <p>Yildiz et al. 1985</p> <p>Greenleaf 1986</p> |
|---|--|
-
- | | |
|---|----------------------|
| <p>63. <i>C. baccatrn</i> var. <i>microcarpunr</i>.
 <i>C. chacoense</i>
 <i>C. frutescens</i> var. <i>cerasiforme</i>
 <i>C. frutescens</i> Tabasco
 — also resistant to CMV</p> | <p>Horvath 1986b</p> |
|---|----------------------|

Tobamoviruses

References

- C. baccalunt* var. *pendulum*
 — resistant to TMV and ToMV
 — also resistant to CMV
- C. pubescens*
 — resistant to ToMV
 also resistant to CMV and PVY
64. *C. annuum* Jangsu (F₁ hybrid), Wongkyo 306 RDA 1986
 also resistant to CMV
 — TMV and CMV resistance derived from
 Taigukgochu, a local variety from Thailand
65. *C. annuum* Agronomico-8 Villalon 1986a
- C. annuum* Tambel-2
 — also resistant to PVY, TEV, and PeMV
66. *C. annuum* Hidalgo Villalon 1986b
 (derived from PI 342947 [resistant to TEV, PVY,
 and TMV] x Avelar [resistant to TEV and PVY])
 — also resistant to PVY, TEV, and PeMV
67. *C. annuum* Hidalgo, Tambel-1, Tambel-2, Tam Mild Villalon 1986c
 Chile-1, Tarn Mild Chile-2, Tam Mild Jalapeño-1
 also resistant to PeMV, PVY, and TEV Villalon et al. 1986
68. *C. annuum* Bangla Green, HS-1, Jamni, Laichi, Longi, Singh and Kaur 1986
 Lorai, Pant C-1, Perennial, S 118-2, S 20-1, Tiwari
 — also resistant to CMV and field resistant to
 leaf curl virus
69. *C. annuum* Catalan, Fado (F₁), Heldor, Jericho, Betti et al. 1986
 Lamuyo, Melody (F₁), Pacific, Rinascita, Shamrock,
 Yolo Wonder B
70. *C. annuum* S 38-2-1, S 38-3-19, S 42-2-4, S 52-1-6, Tewari and Viswanath 1986
 S 96-4-9-3, S 96-4-8, S 101-2-33
 tolerant to TMV
 also tolerant to CMV, PVX, and TLCV
71. *C. annuum* Tans Mild Chile-2 (TMC-2) Villalon et al. 1986
 (derived from Agronomico 8, Avelar [both resistant
 to TEV, PVY, TMV], Cal Compack 528 and Anaheim
 TMR-23)
 — also resistant to PVY, PeMV, and TEV

Tobamoviruses

References

-
72. *C. annuum*. Novi (F₁), Delgado (F₁)
— *L*¹*L*³ alleles for resistance
73. *C. annuum* Tam Rio Grande Gold-Sweet
— also resistant to PeMV, PVY, and TEV
74. *C. chacoense* PI 260429 line Chaton 2-2
— resistant to TMV strains 0, 1, 1.2, 1.2.3
- C. baccatum* var. *pendulum* 152234, 159235
— only local lesions with TMV
75. *C. chinense* PI 315008 (Miscucho)
— resistant to P0, P1, P1.2, but not P1.2.3
76. *C. annuum* Perennial
— immune to TMV and CMV
- C. annuum* CA 1068, G 5, K I, PKV 1
— moderately resistant to TMV and CMV
- C. annuum* Bhivapur 1, Pantnagar 1, Pantnagar 2, PKV 21, Sabouragar, Surjamani
— moderately resistant to TMV
— also resistant to CMV
- C. frutescens* Lawangi
— moderately resistant to CMV
- C. microcarpum*
— hypersensitive resistance to TMV
77. *C. annuum* Punjab Lal
— also resistant to PVY and CMV
78. *C. annuum* PI 152225, PI 159236, PI 315008, Smith SA 411
— resistant to TMV-P (pepper strain)
79. *C. frutescens* Greenleaf Tabasco
— resistant to TMV and ToMV
- C. chinense* PI 152225, PI 159236,
C. frutescens Tabasco McIlhenny
— resistant to TMV, ToMV, and PMMV

Tanzi et al. 1986

Villalon 1987

Pochard 1988
(personal communication)

Pochard 1988

Peshluley and Moghe 1988

Singh and Cheema 1989

Narikawa et al. 1989

AVRDC 1991

Tobaniovirttses

References

80. *C. frutescens* Pusa Sada Bahar
— also tolerant to CMV and leaf curl virus
Tewari 1991

81. *C. annuum* Indonesia line LV 2411,
Thailand line Kee Noo Kao
C. baccatum AVRDC Acc. C 00948
C. baccatum var. *pendulum* AVRDC Acc. C 00755,
C 00773, *C 01178*
C. chinense AVRDC Acc. C 00967, *C 01177*
C. frutescens AVRDC Acc. C 00897, *C 00966*, *C 01610*
— resistant to ToMV and TMV

C. chinense AVRDC Acc. C 00917, *C 01174*
— resistant to ToMV, TMV, and PMMV

C. chinense AVRDC Acc. C 00304
— resistant to ToMV, TMV, and PMMV
— also resistant to PVY

C. baccatum var. *pendulums* AVRDC Acc. C 00753-1, *C 01172*
C. frutescens AVRDC Acc. C 00848
Unknown species AVRDC Acc. C 01090, *C 01175*
— resistant to TMV and ToMV
— also resistant to PVY

C. chinense AVRDC Acc. C 00940
— resistant to ToMV and PMMV
— also resistant to PVY

C. baccatum var. *pendulum* AVRDC Acc. C 00803
Unknown species AVRDC Acc. C 01173
resistant to ToMV
also resistant to PVY

82. *C. annuum* S 20-1, Laichi 2
— also resistant to CMV, PVY, and leaf curl virus
Singh 1992

C. annuum. Tiwari
— also resistant to CMV and leaf curl virus

C. annuum Perennial
— also resistant to CMV, PVY, PVMV, and
leaf curl virus

C. annuum Punjab Lal
— also resistant to CMV, CVMV, and leaf curl virus

Tobamoviruses

References

-
- C. annuum* Bangla Green (BC-1)
 — also resistant to CMV and leaf curl virus
 — partially resistant to PVY
- C. annuum* Lorai
 — also resistant to CMV, PVY, CVMV, and leaf curl virus
83. *C. annuum* Tarn Veracruz
 (derived from *C. annuum*, AC 2207, Avelar, PI 264281, PI 342947, Jalapeno 1158, Jalapeno 100, and Jalapeno L.)
 — resistant to TMV
 — also resistant to CMV, PeMV, PVY, and TRSV
- C. annuum* Avelar
 — resistant to Texas isolate of TMV
 — also resistant to Texas isolates of PVY and TEV
- C. annuum*. PI 342947
 — resistant to Texas isolate of TMV
 — also resistant to Texas isolates of PVY and TEV
84. *C. annuum* AVRDC Acc. C 05342
C. frutescens AVRDC Acc. C 05473
 — resistant to ToMV
 — also resistant to PVY
- C. annuum* AR 89, CNPH 703, G 3
C. frutescens Rawit Kutoaijo
C. baccatum var. *pendulum* PI 260571
 — resistant to ToMV and TMV
- C. annuum* AVRDC Acc. C 05050, C 05063, C 05092, C 05138
C. frutescens AVRDC Acc. C 05419, C 05552
 — resistant to ToMV
85. *C. annuum* PBC 473
 — resistant to ToMV
- C. annuum*. PBC 137, PBC 365, PBC 417, PBC 621, C 04352, Fehrozan, Gregyo
 — resistant to TMV and ToMV
- Villalon et al. 1992
- AVRDC 1993
- AVRDC 1994

Tomato Spotted Wilt VirusReferences

- | | |
|--|---------------------------|
| 1. Several breeding lines | Reifschneider et al. 1986 |
| 2. <i>C. chinense</i> PI 152225, PI 159236
concentric necrotic local lesions only on
inoculated leaves, no systemic infection | Black et al. 1991 |
| 3. <i>C. chinense</i> PI 159236
resistant to 8 Brazilian isolates
(concentric necrotic local lesions on
inoculated leaves)
susceptible to 3 other Brazilian isolates
(systemic necrosis and stunting) | Boiteux and Nagata 1993 |

Unspecified Viruses

References

- | | |
|--|-------------------------|
| 1. PI109252 | Dolan 1957 |
| 2. <i>C. annuum</i> . Jwala
(derived from Puri Red and N.P. 46A)
— resistant to mosaic disease and leaf curl | Tewari 1977, 1982 |
| 3. Some varieties collected from Punjab (India)
resistant to mosaic disease | Akhtar and Ibrahim 1979 |

Appendix I. Viruses reported to infect pepper (Green and Kim 1991)

Taxonomic group	virus	Particle size	Vector	Geographical distribution
FILAMENTOUS VIRUSES				
Potyviruses	Chilli vernal rholde ((VMV)	750 x 12		Asia
	Pepper mild mosaic	714		Venezuela
	Upper motile (PeiAIV)	727		El Salvador, USA
				Thailand
	Pepper severe 1)0SaiC (PoSMV)	761. x 13		Argentina
	Pepper veinal mottle (PV:MV)	770 x 12 850 x 12		West Africa (Ghana, Ivory Coast, Nigeria)
	Peru tomato virus (PTV)	750 x 12		Peru
	Potato virus Y (PVY)	730 x 13		Worldwide
Carlaviruses	Tobacco etch (TEA)	730 x 12-13		USA, Mexico, Sudan
				Nigeria, Venezuela
				(mainly Solanaceae)
Potexviruses	Potato virus M (PVNI)	650 x 12		USSR, India
	Potato virus S (PVS)	650 x 12		USSR, India
Potexviruses	Potato incubi mosa (3\ly)V)	580 x 11-13		Worldwide
	Potato virus X (PVX)	515x)113		Worldwide
ROD-SHAPED VIRUSES				
Tobamoviruses	Pepper mild mottle (POMV)	312 x 18		North America, Australia, Japan, Europe (Denmark, France, Greece, Italy, the Netherlands, Spain)
	Tobacco mosaic (TMV)	310 x 18		wide
	Tomato mosaic (ToMV)	300 x 18		Worldwide
	Shi pepper mottle (ShiPeMV)	3110 x 18		Shi pepper

i WO

Taxonomic grOup	Particle slit)	(leo)i,rapliica I Vector` distribution	!!lost ran
ob,uc mil green nur,.nc (ad(;A9V)		Worldwiile Iermanny, lungnry, Italy, 1JS1\, Brazil, Nurtl1 AI Spain, Canary Islands, Amin Ala)	`olanaceae 1lmbelli(erne Coinmelimicea1 Gesneriaceae
Dnlearnti 'a yellow flee (1)11'2')	00 z 19	Hungary	Solanaceae
obra* cruses	ob,u <<* wine (I RV) onnta eni α iii "1-23z 21-?3 x 183-1 ¹ _{4i}	US[1, I:urt*pr lht/il, Japan	
lii ilt RIP \ IRU;11S			
Lrat ov Host";	P,cet cr eskern ti e lots , (l)bV"YV)	USA, japan	Wide; mainly otvledonac, many vegetables and weeds
Tobacco necrosis and satellite viruses	"Tobacco necrosis (TNN) 21	1\torl Hick	1Nide.
"1'onibusviruses	Tomato bushy stunt (IPSV) 10	U2 \, 1 ur,iPe, North [\iricn	121'2
	i1 cot'an pepper viri;s	ll	urol c, tik rth `clan 1ctir2
1,11)a\ 'Huse;	Broad bean rent (i 1(3 ¹ 2)	Argentina, 4y ^{pe} Europe,] 10an, morocco	Wide; mainly Dlcotyledonac
lepot:nuse,	Iona, c new, spot (I 1lt·2	North America	Wide; moody, herbaceous, ornamentals
	homatl) ring spirt (I on1RV)		Wide; ornamentals, woody, semiwuoclv plants
	Tomato black ring	N	Fiu'opii Wide

continued

Appendix 1. Viruses reported to infect peppers (Green and Kim 1991)					
Isolation group	Virus	Particle size	Isolation	Geographical distribution	Host range
Tomoviruses	Tomato mosaic virus (TMV)	27	B	Europe, USA	Solanaceae
Alfalfa mosaic virus group	Alfalfa mosaic virus (AMV)	27	B	Europe, USA	Solanaceae
Cucumoviruses	Cucumber mosaic virus (CMV)	27	B	Europe, USA	Solanaceae
Pepper mild mottle virus group	Pepper mild mottle virus (PMDV)	27	B	Europe, USA	Solanaceae
Cryptic viruses	Red pepper cryptic virus (RPCV)	30	B	Japan	Capsicoviridae
Tomato spotted wilt virus	Tomato spotted wilt virus (TSWV)	90	B	Worldwide	Wide

Appendix "I, Viruses reported to infect peppers (Green and Kim "1991)

Caiddnon,ic group	Vines	Particle sine	Geographical Vector' distribution	Host range
OT111 k, NOT WELL (TIARACTERIZII) VIRUSI?,S				
ILII ,erne	mosaic		India	N Im,nceac
I rinjal mosaic			IN	Cnl,shrrrrspp.
Chili le ^g curl			(V Sri Lanka, India	Sc,lannceau
Orton l Inb1ndiug			Cuba	CL'/'/lllrr Spy).
LB Cared mosaic.			India	C?/1SICUIN Sri).
				L. i'ac1thrl+lultl
Marigold mottle			N India	Capsicum spp.
Pepper vein bandi/17	17z G79	A	India	C71'iiotrl spp.
Pepper yellow vein mosaic			hngland, l Iun ary, Netherlands	Capsicum spp. 1.i Iurn ;rIran So/armor eillosunl

' A = aphid; B = beetle; C = contact; F = fungus; L = leafhopper; N = nematode; T = thrips; W = whitefly; S = soil, no insect vector found; - = no vector; ? = particle size or vector not identified.

² Only infection by artificial inoculation has been reported.

'For more information on geminiviruses, the reader is referred to the AVRDC Technical Bulletin No. 21: Leaf curl and yellowing viruses of pepper and tomato: an overview.

References

- Abdalla, O.A. and Desjarclins, P.R. 1985. Comparative reaction of resistant and susceptible pepper cultivars to California isolates of poly viruses. *Phytopathology* 75:1349.
- Akhtar, C.M. and Ibrahim, M. 1979. Screening of chilli varieties to mosaic disease. Pakistan Association for the Advancement of Science, 59A-60A.
- Anand, G.P.S., Mishra, M.D., and Singh, A. 1961. Resistance to mosaic in certain chili varieties. *Indian Phytopathology* 14:112-114.
- Annual Rept. Regional Project S-9, Experiment, Georgia (mimeograph).
- Annual Rept. Regional Project NE-9, 1961. Geneva, New York (mimeograph).
- Apablaza, G. and Jorquera, P. 1985. Reactions of sweet pepper (*Capsicum annuum* var. *grossum*) cultivars to the common strain (PVY^c) of potato virus Y under field and greenhouse conditions. *Ciencia e Investigacion Agraria* 12:165-171 (In Spanish).
- AVRDC. 1990a. AVRDC 1988 Progress Report. Asian Vegetable Research and Development Center, Shanhua, Tainan, Taiwan, ROC. 411 p.
- AVRDC. 1990b. AVRDC 1989 Progress Report. Asian Vegetable Research and Development Center, Shanhua, Tainan, Taiwan, ROC. 351 p.
- AVRDC. 1991. AVRDC 1990 Progress Report. Asian Vegetable Research and Development Center, Shanhua, Tainan, Taiwan, ROC. 313 p.
- AVRDC. 1992. AVRDC 1991 Progress Report. Asian Vegetable Research and Development Center, Shanhua, Tainan, Taiwan, ROC. 413 p.
- AVRDC. 1993. AVRDC 1992 Progress Report. Asian Vegetable Research and Development Center, Shanhua, Tainan, Taiwan, ROC. 523 p.
- AVRDC. 1994. AVRDC 1993 Progress Report. Asian Vegetable Research and Development Center, Shanhua, Tainan, Taiwan, ROC. 537 p.
- Balashova, N.N., Timina, O.O., and Rushchuk, V.S. 1983. Genetic basis of breeding vegetable crops for resistance to mosaic. *Tagungsbericht, Akademie der Landwirtschaftswissenschaften der Deutschen Demokratischen Republik* 216:323-330.
- Bansal, R.D., Aulakh, R.K., and Hundal, J.S. 1992. Reaction of different genotypes of pepper (*Capsicum annuum* L.) to cucumber mosaic virus. p. 132-137 In: VIIIth meeting on "Genetics and Breeding on *Capsicum* and Eggplant". 7-10 September 1992, Rome, Italy.
- Barrios, E.P., Mosokar, H.I., and Black, L.L. 1971. Inheritance of resistance to tobacco etch and cucumber mosaic viruses in *Capsicum frutescens*. *Phytopathology* 61:1318.

- Barta, A., Niemi, H.A.T., and Salamon, P. 1985. Establishing TMV resistant pepper varieties. *Capsicum Newsl.* 4:49.
- Berkmortel, van den, L. G. 1977. Breeding pepper for resistance to a strain of TMV. p. 89—92 In: *Capsicum* 77: Report of the Third EUCARPIA Congress on the Genetics and Breeding of Red Pepper. 5—8 July, INRA, Avignon-Montfavet, France. 312 p.
- Bhalla, P.L., Baghel, B.S., and Krishna, A. 1983. Screening of chilli cultivars/lines against leafcurl and anthracnose. p.147. In: Proc. Nat. Seminar Production Technology of Toma to and Chillies, Tamil Nadu Agric. Univ. Coimbatore, India.
- Black, L.L., Hobbs, H.A., and Getti, J.M. 1991. Tomato spotted wilt virus resistance in *Capsicum chinense* PI 15225 and 159236. *Plant Dis.* 75:863.
- Boiteux, L.S. and Nagata, T. 1993. Susceptibility of *Capsicum chinense* PI 159236 to tomato spotted wilt isolates in Brazil. *Plant Dis.* 77:210.
- Boukema, I.W. 1977. Resistance in *Capsicum* to a pepper strain of TMV. p. 85—88 In: *Capsicum* 77, Comptes Rendus du 3^e Congres EUCARPIA sur la Genetique et la Selection du Piment. 5—8 July, INRA, Avignon-Montfavet, France.
- Boukema, I.W. 1978. Resistance in *Capsicum* to a pepper strain of TMV. *Plant Breeding Abstracts* 48(9): 85—88.
- Boukema, I.W. 1980. Allelism of genes controlling resistance to TMV in *Capsicum* L. *Euphytica* 29: 433—439.
- Boukema, I.W. 1982. Resistance to a new strain of TMV in *Capsicum chacoense* Hrmz. *Capsicum Newsl.* 1:49—51.
- Boukema, I.W. 1984. Resistance to TMV in *Capsicum chacoense* Hunz. is governed by an allele of the L-locus. *Capsicum Newsl.* 3:47.
- Boukema, I.W., Jansen, K., and Hofman, K. 1980. Strains of TMV and genes for resistance in *Capsicum*. p. 44—48 In: Proc. Fourth Capsicum EUCARPIA Meeting, Wageningen.
- Chaine-Dogimont, C. 1993. Etude genetique de trois systemes de resistance par hypersensibilite ou sequestration aux trois virus infectant le piment (*Capsicum annuum* L.) These, INA, Paris Grignon, France. 194 p.
- Chew, B.H. 1994. Genetics and breeding for resistance to viruses in hot pepper. In: Chew, B.H., Syed Ralunan, Loke, W.H., and Melor, R. (ed.). Proc. Conf. on Chilli Pepper Production in the Tropics. 13—14 October 1992, Kuala Lumpur, Malaysia (In press).
- Chew, B.H. and Ong, C.A. 1990. Genetics and breeding for chilli veinal mottle and cucumber mosaic virus resistance in hot pepper. p. 55—58. In: Proc. 3rd, Int. Conf. Plant Prot. in the Tropics, Vol. V. 20—23 March 1990, Malaysian Plant Protection Society (MAPPS), Genting Highlands, Malaysia.
- Choi, K.S. and Pae, D.H. 1985. A new hybrid 'Wonkyo 306' with the multi-resistance in *Capsicum annuum*. *Capsicum Newsl.* 4:25—26.
- Cohen, S. 1982. Resistance to CMV transmission by aphids in peppers. La Selection des plantes pour la resistance aux maladies. Colloque Franco-Israelien. 21—16 March 1982, Bordeaux, France. *Agronomie* 59—64.
- Cook, A.A. 1960. Genetics of resistance in *Capsicum annuum* to two virus diseases. *Phytopathology* 50: 364—367.